Totem-Implied Repo

Theory, div-agnostic control and the detection of bad curves

A technical note with proofs, worked examples and a synthetic benchmark

Independent Price Verification – Equity Derivatives

14 October 2026 • Draft for discussion

Abstract

Repo parameters used for independent price verification are implied from Totem: forwards from consensus call and put prices, dividends from the separate Totem dividend service. Because the two services are fed by different contributor panels whose sizes change with tenor, the implied repo curve inherits noise and artefacts from both, and sometimes shows shapes the market does not justify. This note gives (i) an exact decomposition of the implied repo into the option panel's repo and a *panel gap*, with the noise, jump and bias it produces for each mechanism; (ii) a non-identifiability theorem stating that no method using consensus data alone can remove that gap; (iii) a control identity proving that the total of the dividend and repo control lines does not depend on the dividend panel for forward-linear exposure, together with a materiality measure computed from sensitivities alone; (iv) a structural detector built from matched filters and a generalised likelihood-ratio test, calibrated on clean history, that names the mechanism behind each alarm and repairs the curve; and (v) a benchmark on a contributor-level synthetic Totem market comparing it with naive checks and with B-spline plus Gaussian-mixture clustering. Every statement is proved, illustrated by a worked example and checked in the accompanying Python package.

Contents

1 Introduction **4**
- 1.1 Context . . . 4
- 1.2 Questions answered . . . 4
- 1.3 Main results at a glance . . . 4
- 1.4 Reading guide . . . 5

2 Model and notation **5**
- 2.1 Forwards with discrete dividends . . . 5
- 2.2 Carry space . . . 5
- 2.3 The implied repo map . . . 6
- 2.4 Forwards from consensus option prices . . . 7
- 2.5 Dividend buckets and allocation . . . 7
- 2.6 Totem data model . . . 7

3 Anatomy of the implied repo **8**
- 3.1 The panel gap . . . 8
- 3.2 Contributor drop-off . . . 9
- 3.3 Ex-date timing splits . . . 9

3.4 Stale ex-date profile . . . 10
3.5 Dividend bucket errors . . . 10
3.6 Spot snapshot and front dividends . . . 10
3.7 Gross/net definitions and proportional biases . . . 10
3.8 What the data cannot tell . . . 11

4 Repo control from forward and dividend controls 11
4.1 Control lines . . . 11
4.2 The implied-repo Jacobian . . . 12
4.3 The control identity . . . 12
4.4 How much the split matters for a book . . . 13
4.5 Worked examples . . . 13
4.6 When the split does matter . . . 14

5 A statistical model of the Totem–FO carry difference 15
5.1 Observation model . . . 15
5.2 Covariance of consensus means . . . 15
5.3 The option panel . . . 15
5.4 The dividend panel . . . 16
5.5 Dispersion spikes are symptoms, not noise . . . 16

6 Detection: matched filters and the likelihood-ratio test 16
6.1 Templates . . . 16
6.2 Whitening, projection and the matched filter . . . 17
6.3 Family-wise calibration . . . 18
6.4 Attribution, repair and the FO-mismark test . . . 18
6.5 Algorithms . . . 19
6.6 Examples on synthetic names . . . 19
6.7 Small neural networks are matched filters . . . 20

7 Benchmark on a synthetic Totem market 22
7.1 Design . . . 22
7.2 Results . . . 23
7.3 Reading the results . . . 23
7.4 Limits of the evidence . . . 26

8 Where B-splines, Gaussian mixtures and neural networks fit 26
8.1 What B-spline coefficients cannot see . . . 26
8.2 Why a mixture on the difference flags FO mismarks . . . 27
8.3 Where they help . . . 27

9 Implementation and operations 27
9.1 Package . . . 27
9.2 Production inputs . . . 28
9.3 Monthly procedure . . . 28

9.4 Calibration and governance 29

10 Conclusion **29**

A Notation **29**

B Additional proofs and remarks **29**

C Synthetic market specification **30**

D Reproducibility **30**

1 Introduction

1.1 Context

An independent price verification (IPV) function controls the parameters that front office (FO) uses to value and risk-manage equity derivatives. For repo, the reference is not observed directly. It is *implied*: Totem publishes consensus prices of calls and puts, from which forwards are extracted by put–call parity, and a separate Totem service publishes consensus dividends. The repo curve that reconciles the two is then compared with FO's repo marks, and the differences are multiplied by FO sensitivities to produce control lines (variances, reserves, prudent-valuation adjustments).

Two features of Totem make this delicate. The option service and the dividend service are fed by different contributors, and within each service the number of contributors changes with tenor; a typical single stock might have fifteen contributors at one year on options and ten on dividends, fewer beyond two years. The implied repo mixes the two consensus means, and so inherits the noise of both and the artefacts of their mismatch. Controllers see this as irregular shapes and unjustified jumps in implied repo curves.

1.2 Questions answered

This note answers four questions:

(a) **Where do irregular implied repo curves come from?** Section 3 decomposes the implied repo exactly into the option panel's repo plus a panel gap, and derives the signature of each mechanism (panel gap, contributor drop-off, ex-date timing, stale ex-date profile, dividend bucket error, spot snapshot, gross/net definition).

(b) **Can a better statistical method recover the "true" repo?** No: Theorem 3.12 shows that consensus data cannot identify the split between repo and dividends of the option panel.

(c) **Do we need a clean Totem repo curve to control repo?** For forward-linear exposure, no: Theorems 4.4 and 4.5 prove that the total of the dividend and repo lines does not depend on the dividend panel, and Proposition 4.8 measures, from sensitivities alone, how much the split matters for a given book.

(d) **How should bad curves be detected?** Sections 5 and 6 build a structural detector: a noise model computed from Totem's own statistics, one matched-filter template per mechanism, a calibrated false-alarm rate and an explanation in natural units for each alarm. Section 7 benchmarks it, and Section 8 explains where B-spline coefficients, Gaussian mixtures and small neural networks fit.

1.3 Main results at a glance

Insight

- To first order, $b_{\text{implied}} = \bar{b}_{\mathcal{A}} + (\overline{D}_{\mathcal{A}} - \overline{D}_{\mathcal{B}})/(T(S-D))$ (Proposition 3.1). The gap alone gives 6.5 bp of repo noise at one year and 51.6 bp at five years for a typical single stock.
- The mechanisms have distinct signatures in *carry space* $B = bT$: a constant (spot), a step (dividend amount, panel change), a spike (forward outlier, ex-date timing), a seasonal pattern (allocation profile).
- For forward-linear books, the full-revaluation Totem P&L is *exactly* independent of the dividend panel when expiries sit on Totem pillars (Theorem 4.5); the line decomposition reproduces this to first order (Theorem 4.4). The dividend panel only matters through the non-forward exposure $g = s_D + s_b J$.

- The detector's test statistics are exact matched filters (Theorem 6.1). One variance scale per category, calibrated on clean history, sets the false-alarm rate exactly (Proposition 6.5).
- In the benchmark (5400 synthetic test names, every detector at 2% false alarms), the structural detector and a Gaussian mixture on B-spline coefficients of the Totem–FO difference detect a similar share of anomalies on average (73% and 73%). The structural detector also finds forward outliers (97% against at most 87%), explains 88% of its alarms correctly, and does not mistake genuine FO mismarks for bad Totem curves (3% against 75%).

1.4 Reading guide

Section 2 fixes notation and proves the elementary facts the rest relies on (bootstrap, parity, allocation). Section 3 is the diagnosis; Section 4 the control identity; Sections 5 and 6 the detector; Section 7 the evidence; Section 8 the comparison with other approaches; Section 9 the implementation. Proofs follow each statement; longer derivations are in Appendix B. Every numerical example is produced by the code in the repository (Appendix D).

2 Model and notation

2.1 Forwards with discrete dividends

Time is measured in years from the valuation date. Let S be the spot, $r(t)$ the continuously compounded zero rate used for discounting and funding, and $b(t)$ the zero *repo* (borrow cost) curve, so that holding the stock financed by a repo earns $r - b$. Write $R(t) = r(t)\,t$, $B(t) = b(t)\,t$ and $G(t) = e^{R(t)-B(t)}$. Cash dividends d_i go ex at times t_i.

Definition 2.1 (Forward)**.** The forward for delivery at T is

$$F(T) = G(T)(S - D(T)), \qquad D(T) = \sum_{t_i \le T} \frac{d_i}{G(t_i)}. \tag{1}$$

$D(T)$ is the *carry-discounted cumulative dividend*: dividends discounted at the carry rate $r - b$.

Equation (1) is the usual replication argument: buy the stock financed at $r - b$, pay the dividends forward. With flat curves it reads $F(T) = Se^{(r-b)T} - \sum_{t_i \le T} d_i e^{(r-b)(T-t_i)}$. Note that D depends on the repo curve through $G(t_i)$; this coupling is weak (a few basis points of repo per year on the dividend discounting) but the bootstrap below handles it exactly.

2.2 Carry space

Definition 2.2 (Carry)**.** The *cumulative repo* or *carry* is $B(T) = b(T)\,T$. For a quantity observed at pillars $T_1 < \cdots < T_K$ we write $B_k = B(T_k)$, $b_k = b(T_k)$, $D_k = D(T_k)$, $F_k = F(T_k)$.

Lemma 2.3 (Additivity in carry)**.** $\ln F(T) = R(T) - B(T) + \ln(S - D(T))$. *Consequently, perturbations of the forward, the dividends and the carry satisfy, to first order,*

$$\delta B(T) = -\,\delta \ln F(T) - \frac{\delta D(T)}{S - D(T)} + \delta R(T). \tag{2}$$

Proof. Take logarithms in Eq. (1); differentiate, using $\delta \ln(S - D) = -\delta D/(S - D) + O(\delta D^2)$. □

Insight

In carry space the mechanisms add up and nothing is amplified. A carry error ε at tenor T is a repo error ε/T: the same 0.1% error in the forward is 120 bp of repo at one month and 2 bp at five years. This is why implied repo curves look so noisy at the short end, and why all tests in this note work on B, not on b.

2.3 The implied repo map

Given forwards $F_1, \dots, F_K$ at the pillars, the rate curve and the dividends, the *implied repo* is the repo curve, piecewise linear in B (flat forward repo between pillars, $B(0) = 0$), that reprices the forwards. It is solved pillar by pillar. Fix $B_1, \dots, B_{k-1}$ and let $x = B_k$. For a dividend between the previous pillar and T_k, $B(t_i) = B_{k-1} + w_i(x - B_{k-1})$ with $w_i = (t_i - T_{k-1})/(T_k - T_{k-1}) \in (0, 1]$, so the model forward at T_k is

$$\Phi_k(x) = Se^{R_k - x} - \sum_{t_i \le T_{k-1}} d_i \, e^{(R_k - R_i) - (x - B(t_i))} - \sum_{T_{k-1} < t_i \le T_k} d_i \, e^{(R_k - R_i) - (1 - w_i)(x - B_{k-1})}. \tag{3}$$

Lemma 2.4 (Monotonicity)**.** *$\Phi_k'(x) \le -\Phi_k(x)$ for all x. In particular Φ_k is strictly decreasing wherever the forward is positive, and $g_k(x) = \ln \Phi_k(x) - \ln F_k$ satisfies $g_k'(x) \le -1$.*

Proof. Differentiating Eq. (3) term by term, $\Phi_k'(x) = -Se^{R_k - x} + \sum_{t_i \le T_{k-1}} d_i e^{(\cdot)} + \sum_{T_{k-1} < t_i \le T_k} (1 - w_i)\, d_i e^{(\cdot)}$. Every dividend term is positive and $1 - w_i \le 1$, so replacing $(1 - w_i)$ by 1 can only increase the right-hand side, which then equals $-\Phi_k(x)$. Dividing by $\Phi_k > 0$ gives $g_k' \le -1$. □

Proposition 2.5 (Well-posed bootstrap)**.** *Assume $S - D(T_{k-1}) > 0$ for the dividends already solved. Then for every $F_k > 0$ there is a unique B_k with $\Phi_k(B_k) = F_k$ and $\Phi_k > 0$ on the solution branch. Newton's method on g_k converges locally quadratically, and globally when safeguarded by bisection.*

Proof. As $x \to -\infty$,

$$e^x \Phi_k(x) \to e^{R_k}\Big(S - \sum_{t_i \le T_{k-1}} d_i e^{-R_i + B(t_i)}\Big) = e^{R_k}\big(S - D(T_{k-1})\big) > 0,$$

because the in-period terms grow only like $e^{-(1-w_i)x}$; hence $\Phi_k \to +\infty$. As x increases, Φ_k decreases strictly while positive (Lemma 2.4) and either tends to 0 or crosses zero, so by continuity it takes every value in $(0, \infty)$ exactly once on the branch where it is positive. On that branch g_k is C^2 with $|g_k'| \ge 1$, so the Newton iteration $x \mapsto x - g_k(x)/g_k'(x)$ converges quadratically near the root; the bisection safeguard (keep a bracket $[\ell, u]$ with $g_k(\ell) > 0 > g_k(u)$ and fall back to its midpoint) makes convergence global. □

Example 2.6 (A bootstrap by hand)**.** Spot 100, flat rate 3%, dividends 1.5 at $t = 0.30$ and 2.0 at $t = 0.80$, pillars 0.25, 0.5, 1, true repo 80, 70, 60 bp. The forwards and the implied repo are

Pillar T_k	Forward F_k	True b_k (bp)	Implied b_k (bp)
0.25	100.5515	80.00	80.00
0.50	99.6494	70.00	70.00
1.00	98.8928	60.00	60.00

At the third pillar the Newton residual $|g_3|$ falls from 3.1×10^{-5} to 3.5×10^{-12} in one step and to zero in the next. Without dividends g_k is linear and a single step suffices.

2.4 Forwards from consensus option prices

Lemma 2.7 (Parity and the consensus forward)**.** *Let each contributor j quote European calls and puts with a common discount factor DF. Then*

(i) $C_j(K) - P_j(K) = DF\,(F_j - K)$ *for every strike;*

(ii) if the consensus averages the same contributors at every strike, $\overline{C}(K) - \overline{P}(K) = DF\,(\overline{F} - K)$ *with* $\overline{F}$ *the mean of the contributors' forwards;*

(iii) the strike K^* *where* $\overline{C} = \overline{P}$ *equals* $\overline{F}$ *and does not depend on DF; multiplying all prices by a common factor leaves* K^* *unchanged.*

Proof. (i) is put–call parity [2]. (ii) follows by averaging (i), which is linear. (iii) Set (ii) to zero; scaling both prices scales the line $K \mapsto \overline{C} - \overline{P}$ without moving its root. □

Example 2.8. With strikes 90, 100, 110, puts 2.1, 5.4, 10.9 and calls built from $F = 102.3$, $DF = 0.97$, the weighted regression of $\overline{C} - \overline{P}$ on K returns $F = 102.30$ and $DF = 0.9700$. Scaling all prices by 1.02 (a wrong discount factor) returns $F = 102.30$ and $DF = 0.9894$: the forward is unchanged.

Remark 2.9. Two cases break (ii)–(iii): outlier screening applied separately per strike and per option type (the averaged contributors differ between calls and puts), and American exercise (parity becomes an inequality). The fitted DF compared with the curve gives an implied funding spread that flags such expiries. The controller keeps its own rate curve on both sides of the comparison, so rates cancel in the Totem–FO difference.

2.5 Dividend buckets and allocation

Definition 2.10 (Allocation)**.** Totem dividend levels L_y are sums of dividends going ex in period $y = (e_{y-1}, e_y]$ (typically calendar years). A profile assigns weights $\pi_i \geq 0$ with $\sum_{i \in y} \pi_i = 1$ to the ex-dates of each period; the allocated dividends are $d_i = L_y \pi_i$. The controller normally takes the FO ex-date calendar as profile.

Lemma 2.11. *Holding G fixed,* $D_k = \sum_y A_{ky} L_y$ *with* $A_{ky} = \sum_{i \in y,\, t_i \leq T_k} \pi_i / G(t_i)$.

Proof. Substitute $d_i = L_y \pi_i$ in Eq. (1) and group by period. □

A_{ky} is the carry-discounted share of period y that goes ex before expiry k. It is 0 or close to $1/G$ except for the periods that straddle an expiry, which is exactly where the profile matters.

2.6 Totem data model

Assumption 2.12 (Consistent contributors)**.** Contributor j holds views $b_j(\cdot)$ and $D_j(\cdot)$ and quotes option prices consistent with $F_j(T) = G_j(T)(S - D_j(T))$, with the common rate curve and spot.

Assumption 2.13 (Panels)**.** At expiry T_k the option panel $\mathcal{A}_k$ has $n_{A,k}$ contributors; for period y the dividend panel $\mathcal{B}_y$ has $n_{B,y}$. Panels are nested in tenor ($\mathcal{A}_{k+1} \subseteq \mathcal{A}_k$, $\mathcal{B}_{y+1} \subseteq \mathcal{B}_y$). View errors are independent across contributors. The two services share m contributors ($m = 0$ for disjoint panels).

Assumption 2.14 (Consensus)**.** The consensus is the arithmetic mean over the panel. Totem reports, per expiry and per period, the consensus, the dispersion across contributors and their number.

Definition 2.15 (Totem-implied repo and carry difference)**.** The Totem-implied repo b^{T} is the bootstrap of the consensus forwards $\overline{F}_k$ with the allocated consensus dividends $\overline{L}_y$. With the FO repo b^{FO} at the same pillars, the *carry difference* is

$$y_k = B_k^{\mathrm{T}} - B_k^{\mathrm{FO}} = (b_k^{\mathrm{T}} - b_k^{\mathrm{FO}})\, T_k, \qquad k = 1, \dots, K. \tag{4}$$

3 Anatomy of the implied repo

3.1 The panel gap

For a pillar T write $\bar{b}_{\mathcal{A}}(T)$ and $\overline{D}_{\mathcal{A}}(T)$ for the averages of the repo and dividend views over the option panel, and $\overline{D}_{\mathcal{B}}(T)$ for the dividend consensus of the dividend panel after allocation. Views are perturbations of a common centre, $b_j = b_0 + \delta b_j$ and $D_j = D_0 + \delta D_j$, of order δ.

Proposition 3.1 (Panel-gap decomposition)**.** *Under Assumptions 2.12 to 2.14,*

$$b^{\mathrm{T}}(T) = \bar{b}_{\mathcal{A}}(T) + \frac{\overline{D}_{\mathcal{A}}(T) - \overline{D}_{\mathcal{B}}(T)}{T\,(S - D_0(T))} + O(\delta^2). \tag{5}$$

Proof. For $j \in \mathcal{A}$, Lemma 2.3 gives $\ln F_j = \ln F_0 - T\,\delta b_j - \delta D_j/(S - D_0) + O(\delta^2)$. The consensus forward is the mean of the F_j, and ln of a mean equals the mean of the ln up to a variance term of order δ^2, so $\ln \overline{F} = \ln F_0 - T\,\overline{\delta b}_{\mathcal{A}} - \overline{\delta D}_{\mathcal{A}}/(S - D_0) + O(\delta^2)$. The controller allocates the dividend consensus, $\overline{D}_{\mathcal{B}} = D_0 + \overline{\delta D}_{\mathcal{B}}$, and bootstraps: $B^{\mathrm{T}} = R - \ln \overline{F} + \ln(S - \overline{D}_{\mathcal{B}}) = B_0 + T\,\overline{\delta b}_{\mathcal{A}} + (\overline{\delta D}_{\mathcal{A}} - \overline{\delta D}_{\mathcal{B}})/(S - D_0) + O(\delta^2)$. Divide by T and use $b_0 + \overline{\delta b}_{\mathcal{A}} = \bar{b}_{\mathcal{A}}$. □

The implied repo is the option panel's own repo plus a *panel gap*: the difference between the option contributors' dividend view, which is never observed, and the dividend panel's, divided by $T(S - D)$. For a single contributor the forward and dividend errors cancel in the implied repo; across two panels they do not.

Corollary 3.2 (Size of the gap)**.** *Let the dividend view errors at T be i.i.d. with variance $\sigma_D^2(T)$ in both populations, and let the two panels share m contributors who use the same dividend view in both services. Then*

$$\mathrm{sd}[b^{\mathrm{T}} - \bar{b}_{\mathcal{A}}] = \frac{\sigma_D(T)}{T\,(S - D(T))} \sqrt{\frac{1}{n_A} + \frac{1}{n_B} - \frac{2m}{n_A n_B}}. \tag{6}$$

Proof. $\overline{D}_{\mathcal{A}} - \overline{D}_{\mathcal{B}} = \sum_j (a_j - c_j)\,\delta D_j$ with $a_j = 1/n_A$ on $\mathcal{A}$, $c_j = 1/n_B$ on $\mathcal{B}$ and zero elsewhere. Its variance is $\sigma_D^2 \sum_j (a_j - c_j)^2 = \sigma_D^2 [m(1/n_A - 1/n_B)^2 + (n_A - m)/n_A^2 + (n_B - m)/n_B^2]$, which simplifies to $\sigma_D^2\,(1/n_A + 1/n_B - 2m/(n_A n_B))$. □

With identical panels ($m = n_A = n_B$) the gap vanishes; with disjoint panels it is largest. The dispersion $\sigma_D(T)$ of a cumulative dividend grows with T, roughly like T or faster, so the gap is small in carry at the short end but does not shrink in repo at the long end.

Example 3.3 (Size of the gap for a single stock)**.** Spot 100, dividend yield 4%, contributor dispersion $1\% + 3\% \cdot T$ of the cumulative dividend, panel sizes of the synthetic EU-stock category. The closed form (6) with $m = 0$ and a contributor-level simulation with no repo disagreement give

Tenor	$D(T)$	$\sigma_D(T)$	n_A	n_B	Formula (bp)	Simulation (bp)
3M	1.0	0.02	16	12	2.7	2.7
6M	2.0	0.05	16	12	3.9	3.9
1Y	4.0	0.16	15	12	6.5	6.5
2Y	8.0	0.56	12	10	13.0	12.7
3Y	12.0	1.20	7	7	24.3	24.5
5Y	20.0	3.20	6	4	51.6	52.3

At one year, if six of the contributors were common to both services ($m = 6$, $n_A = 15$, $n_B = 12$), the gap would fall from 6.5 to 4.8 bp.

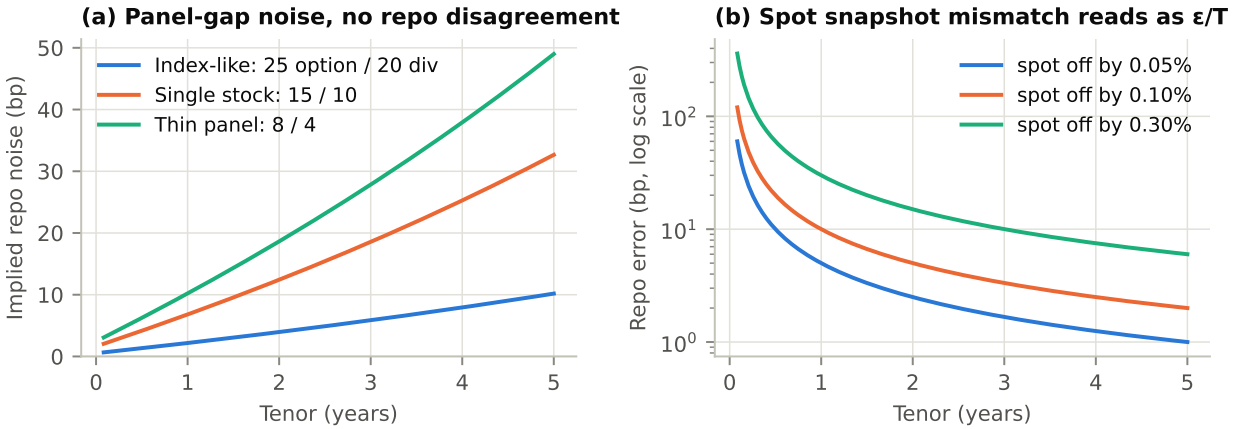


Figure 1: (a) Repo noise from the panel gap alone, Eq. (6), for three panel configurations. (b) A spot snapshot mismatch ε is constant in carry and therefore reads as ε/T in repo (Proposition 3.10).

3.2 Contributor drop-off

Proposition 3.4 (Means of nested panels)**.** *Let contributor j's error at tenor k be e_{jk}, independent across contributors, with* $\operatorname{Var} e_{jk} = \sigma_k^2$ *and* $\operatorname{Cov}(e_{jk}, e_{jl}) = \rho_{kl}\sigma_k\sigma_l$*. If the panels are nested with sizes n_k, the panel means $\bar{e}_k$ satisfy*

$$\operatorname{Cov}(\bar{e}_k, \bar{e}_l) = \frac{\rho_{kl}\,\sigma_k\sigma_l}{\max(n_k, n_l)}. \tag{7}$$

In particular, with persistent views ($\rho = 1$, $\sigma_k = \sigma_l = \sigma$) and $n_l \le n_k$, $\operatorname{sd}(\bar{e}_l - \bar{e}_k) = \sigma\sqrt{1/n_l - 1/n_k}$.

Proof. $\operatorname{Cov}(\bar{e}_k, \bar{e}_l) = \frac{1}{n_k n_l}\sum_{j \le \min(n_k, n_l)} \rho_{kl}\sigma_k\sigma_l = \frac{\min(n_k, n_l)}{n_k n_l}\rho_{kl}\sigma_k\sigma_l$, and $\min/(n_k n_l) = 1/\max$. Then $\operatorname{Var}(\bar{e}_l - \bar{e}_k) = \sigma^2(1/n_l + 1/n_k - 2/n_k) = \sigma^2(1/n_l - 1/n_k)$. □

Example 3.5. In units of the contributor dispersion, the consensus jumps by itself by $0.14\,\sigma$ when a panel shrinks from 16 to 12, $0.24\,\sigma$ from 12 to 7, $0.28\,\sigma$ from 15 to 7 and $0.39\,\sigma$ from 10 to 4. A jump of that size at a panel drop is noise, not information; the noise model of Section 5 reproduces it through Eq. (7).

3.3 Ex-date timing splits

Proposition 3.6 (Timing split)**.** *Let a dividend d_i go ex at $t_i \in (T_k, T_{k+1})$ in the dividend allocation, and let a fraction p of the option panel place it before T_k (all place it before T_{k+1}). Writing $\tilde{d}_i = d_i/G(t_i)$, to first order*

(i) the carry difference moves at T_k only: $\delta y_k = p\,\tilde{d}_i/(S - D_k)$, i.e. a repo spike $p\,\tilde{d}_i/(T_k(S - D_k))$;

(ii) in forward repo between pillars this is a dipole: $+\delta y_k/(T_k - T_{k-1})$ then $-\delta y_k/(T_{k+1} - T_k)$;

(iii) the dispersion of contributor log-forwards at T_k rises by $p(1-p)\,\tilde{d}_i^2/(S - D_k)^2$ in variance.

Proof. (i) At T_k a fraction p of the option panel includes the dividend, so $\ln \overline{F}_k$ falls by $p\,\tilde{d}_i/(S - D_k)$ (Lemma 2.3) while the allocated D_k^{T} excludes it; by Eq. (2) B_k^{T} rises by the same amount. Before T_k nobody includes the dividend, after T_{k+1} everybody does, so no other pillar moves. (ii) Forward repo is the slope of B between pillars. (iii) Law of total variance: the contributors form two groups whose mean log-forwards differ by $\tilde{d}_i/(S - D_k)$, with weights p and $1 - p$. □

Example 3.7. In the gallery example of Fig. 4(c) a dividend of 2.18 goes ex 17 days after the six-month expiry and 37% of the option panel places it before. The spike is $0.37\times2.18/(0.5\times98)\approx1.6\%$, about 165 bp of repo at one pillar. The contributor dispersion at that expiry is almost ten times that of its neighbours, consistent with (iii).

3.4 Stale ex-date profile

Proposition 3.8 (Profile shift)**.** *Let the option contributors use the true ex-dates t_i while the allocation profile places them at $t_i+\Delta$. Then, for $\Delta>0$ (symmetrically for $\Delta<0$),*

$$\delta y_k = \sum_{i:\,T_k\in[t_i,\,t_i+\Delta)} \frac{\tilde d_i}{S-D_k}\,:$$

only pillars lying between a true ex-date and its shifted copy move.

Proof. $D_k^{\mathrm T}-D_k=\sum_i \tilde d_i(\mathbf{1}\{t_i+\Delta\le T_k\}-\mathbf{1}\{t_i\le T_k\})$, which is $-\tilde d_i$ exactly when $t_i\le T_k<t_i+\Delta$; apply Eq. (2). □

For seasonal payers the affected pillars recur every year (for example the April expiries of European stocks), which produces the zigzag that controllers recognise. Period ends are never affected, because the period totals are right.

3.5 Dividend bucket errors

Proposition 3.9 (Bucket error)**.** *If the Totem level of period y is off by δL, then $\delta y_k=-A_{ky}\,\delta L/(S-D_k)$. For a period with a single ex-date this is a step starting at the first pillar after the ex-date; in repo it is a jump that decays like $1/T$.*

Proof. Lemma 2.11 gives $\delta D_k=A_{ky}\delta L$; apply Eq. (2) with the forwards unchanged. A_{ky} is zero before the ex-date and constant ($\approx 1/G(t_i)$) after. □

3.6 Spot snapshot and front dividends

Proposition 3.10 (Spot snapshot)**.** *If option contributors price off a spot $S(1+\varepsilon)$ while the controller uses S, then*

$$y_k=-\ln\Big(1+\frac{\varepsilon S}{S-D_k}\Big)\approx-\frac{\varepsilon S}{S-D_k}\qquad \text{for every } k, \tag{8}$$

nearly constant across pillars (exactly constant without dividends), hence a repo error close to $-\varepsilon/T_k$. A dividend before the first pillar mis-stated by δd produces the same signature, $-\delta\tilde d/(S-D_k)$.

Proof. Each contributor's forward becomes $G(S(1+\varepsilon)-D)$, so $\ln\overline{F}_k$ moves by $\ln((S(1+\varepsilon)-D_k)/(S-D_k))$; by Eq. (2) $B_k^{\mathrm T}$ moves by the opposite. For the front dividend, D_k moves by $\delta\tilde d$ at every pillar. □

3.7 Gross/net definitions and proportional biases

Proposition 3.11 (Proportional dividend bias)**.** *If every Totem dividend is $(1+\tau)$ times the dividend embedded in the forwards (for example gross against net of withholding tax), then $y_k\approx-\tau D_k/(S-D_k)$. For a continuous dividend yield q, $D_k\approx SqT_k$ and $y_k\approx-\tau q\,T_k$, which has exactly the shape of a repo level difference of $-\tau q$.*

Proof. $\delta D_k=\tau D_k$ in Eq. (2); for a yield, $D_k=S(1-e^{-qT_k})\approx SqT_k$ and $S-D_k\approx S$. □

3.8 What the data cannot tell

Theorem 3.12 (Non-identifiability of the split)**.** *Suppose European option prices depend on a contributor's views only through its forwards. For any contributor-level transformation* $(b_j, D_j) \mapsto (b_j + \delta_j, D'_j)$ *with* $\delta_j(T)\,T = \ln(S - D'_j(T)) - \ln(S - D_j(T))$*, every option price, and hence every option-panel statistic (consensus, dispersion, count), is unchanged; the dividend-panel statistics are unchanged too. The implied repo* b^{T}*, a function of these statistics, is therefore unchanged, while the option panel's repo* $\bar{b}_{\mathcal{A}}$ *moves by the average of the* δ_j*. Consequently the panel gap* $b^{\mathrm{T}} - \bar{b}_{\mathcal{A}}$ *is not a function of the observable data.*

Proof. The transformation keeps $\ln F_j = R - B_j + \ln(S - D_j)$ fixed for every j and every T, so each contributor's option prices are unchanged and so is every statistic computed from them. The dividend panel is untouched. b^{T} is computed from these statistics only, so it is unchanged, whereas $\bar{b}_{\mathcal{A}}$ shifts by the mean of the δ_j, which can be any value. □

Corollary 3.13. *No estimator built from Totem consensus data alone (curve smoothing, clustering, neural networks) can remove the panel gap. What remains possible is to model its size (Section 5), to recognise the structured mechanisms by their signature (Section 6), and to avoid needing the split where the exposure does not (Section 4).*

Remark 3.14 (What breaks the degeneracy). American options carry some information on the split through early exercise; overlapping contributors reduce the gap (Corollary 3.2); external anchors pin either side directly: listed dividend futures, total-return futures that quote a financing spread, securities-lending fees for the short end of single stocks.

Insight

Summary of signatures in carry space (the detector's templates in brackets):

Mechanism	Carry difference y	Repo difference
Panel gap (3.1)	smooth noise plus increments at ex-dates	noise roughly flat in tenor
Drop-off (3.4)	step at the drop [step]	jump decaying as $1/T$
Timing split (3.6)	one pillar [spike]	spike, dipole in forward repo
Stale profile (3.8)	in-season pillars [season, spike]	zigzag
Bucket error (3.9)	step after the ex-date [step]	jump decaying as $1/T$
Spot snapshot (3.10)	constant [level]	hyperbola ε/T
Proportional bias (3.11)	ramp of steps; $\approx \beta T$ for a yield	steps; a level for an index
FO mismark	smooth $\delta b(T)\,T$ [null model]	smooth level or slope

4 Repo control from forward and dividend controls

4.1 Control lines

Collect the repo zero rates at the pillars and the dividend levels per period in $\theta = (b, L) \in \mathbb{R}^K \times \mathbb{R}^Y$. The spot, the rate curve and the allocation profile are fixed, so the pillar forwards are a smooth map $F(\theta) \in \mathbb{R}^K$. By Proposition 2.5 the *implied repo map* $\beta(F, L)$, the unique b with $F(b, L) = F$, is well defined.

FO marks are $\theta^{\mathrm{FO}} = (b^{\mathrm{FO}}, L^{\mathrm{FO}})$. Totem gives forwards F^{T} and dividend levels L^{T}, hence $b^{\mathrm{T}} = \beta(F^{\mathrm{T}}, L^{\mathrm{T}})$. A position has value $V(\theta)$ with FO sensitivities $s_b = \partial_b V(\theta^{\mathrm{FO}}) \in \mathbb{R}^K$ and $s_D = \partial_L V(\theta^{\mathrm{FO}}) \in \mathbb{R}^Y$. Write $\Delta L = L^{\mathrm{T}} - L^{\mathrm{FO}}$, $\Delta b = b^{\mathrm{T}} - b^{\mathrm{FO}}$ and $\Delta b^F = \beta(F^{\mathrm{T}}, L^{\mathrm{FO}}) - b^{\mathrm{FO}}$: the repo difference implied by Totem forwards *with FO dividends*.

Definition 4.1 (Control lines)**.**

$$\text{div line} = s_D \cdot \Delta L, \qquad \text{repo line} = s_b \cdot \Delta b, \qquad \text{forward line} = s_b \cdot \Delta b^F,$$

and the *total* is the sum of the dividend and repo lines.

Definition 4.2 (Forward-linear exposure)**.** A position is forward-linear if $V(\theta) = \Phi\big(F(\theta)\big)$ for a C^2 function Φ: its value depends on repo and dividends only through the forwards (European options with a vol surface that does not depend on them, forwards, futures).

4.2 The implied-repo Jacobian

Lemma 4.3. *$J := \partial_L \beta(F, L) = -(\partial_b F)^{-1} \partial_L F$. The matrix $\partial_b F$ is lower triangular with diagonal entries $\partial F_k / \partial b_k \le -T_k F_k < 0$, so J exists. To leading order*

$$J_{ky} \approx -\frac{A_{ky}}{T_k\,(S - D_k)}. \tag{9}$$

Proof. Differentiate $F(\beta(F, L), L) = F$ in L: $\partial_b F\, J + \partial_L F = 0$. With flat forward repo between pillars, F_k depends on $b_1, \dots, b_k$ only, so $\partial_b F$ is lower triangular; its diagonal is $T_k\, \Phi_k'(B_k) \le -T_k F_k$ by Lemma 2.4. For the leading order, hold the forwards fixed in Eq. (2): $\delta B_k = -\delta D_k/(S - D_k) = -A_{ky}\delta L_y/(S - D_k)$, and divide by T_k. □

4.3 The control identity

Theorem 4.4 (Div-agnostic total)**.** *For any position,*

$$\textit{div line} + \textit{repo line} = \textit{forward line} + g \cdot \Delta L + O(\|\Delta\|^2), \qquad g = s_D + s_b\, J, \tag{10}$$

with J evaluated at $(F^{\mathrm{T}}, L^{\mathrm{FO}})$ and $\|\Delta\| = \|\Delta L\| + \|\Delta b^F\|$. If the position is forward-linear, then $g = O(\|\Delta\|)$, so the total equals the forward line up to second order: it does not depend on the dividend panel to first order.

Proof. A first-order expansion of β in L gives $\beta(F^{\mathrm{T}}, L^{\mathrm{T}}) = \beta(F^{\mathrm{T}}, L^{\mathrm{FO}}) + J\Delta L + O(\|\Delta L\|^2)$, i.e. $\Delta b = \Delta b^F + J\Delta L + O(\|\Delta L\|^2)$. Hence $s_D \cdot \Delta L + s_b \cdot \Delta b = s_b \cdot \Delta b^F + (s_D + s_b J) \cdot \Delta L + O(\|\Delta\|^2)$. For $V = \Phi \circ F$, $s_b = \nabla\Phi\, \partial_b F(\theta^{\mathrm{FO}})$ and $s_D = \nabla\Phi\, \partial_L F(\theta^{\mathrm{FO}})$, so $g = \nabla\Phi\, \big[\partial_L F + \partial_b F\, J\big](\theta^{\mathrm{FO}})$. At $\theta' = (\beta(F^{\mathrm{T}}, L^{\mathrm{FO}}), L^{\mathrm{FO}})$ the bracket vanishes by Lemma 4.3; since $\theta' - \theta^{\mathrm{FO}} = (\Delta b^F, 0)$ and the derivatives are Lipschitz, the bracket at θ^{FO} is $O(\|\Delta b^F\|)$, hence $g \cdot \Delta L = O(\|\Delta\|^2)$. □

Theorem 4.5 (Exactness of full revaluation)**.** *If the position is forward-linear and all its expiries are Totem pillars, the full-revaluation control $V(\theta^{\mathrm{T}}) - V(\theta^{\mathrm{FO}}) = \Phi(F^{\mathrm{T}}) - \Phi\big(F(\theta^{\mathrm{FO}})\big)$ does not depend on the Totem dividends L^{T} at all.*

Proof. By construction of the implied repo, $F(\theta^{\mathrm{T}}) = F\big(\beta(F^{\mathrm{T}}, L^{\mathrm{T}}), L^{\mathrm{T}}\big) = F^{\mathrm{T}}$ whatever L^{T} is. □

Remark 4.6 (Expiries between pillars). For an expiry between two pillars the forward depends on the interpolated carry and on the allocated dividends in between, so a residual dependence on L^{T} remains. It is of the size of the dividends going ex between the expiry and its neighbouring pillars, exactly the timing and profile effects of Propositions 3.6 and 3.8, which the detector watches.

Proposition 4.7 (Second-order term)**.** *For one pillar, $F(b, D) = e^{R - bT}(S - D)$ and $V = \Phi \circ F$, the first-order total equals*

$$\Phi'(F)\,\Delta F - \Phi'(F)\,F\Big[\tfrac{1}{2}T^2\Delta b^2 + \frac{T\,\Delta b\,\Delta D}{S - D}\Big] + O(\|\Delta\|^3), \tag{11}$$

whereas the exact control is $\Phi'(F)\Delta F + \frac{1}{2}\Phi''(F)\Delta F^2 + O(\|\Delta\|^3)$. The dependence of the first-order total on the dividend panel is therefore entirely in the bracket.

Proof. Expand F to second order: $F_b = -TF$, $F_D = -F/(S-D)$, $F_{bb} = T^2F$, $F_{bD} = TF/(S-D)$, $F_{DD} = 0$, so $\Delta F = F_b\Delta b + F_D\Delta D + \frac{1}{2}T^2F\Delta b^2 + TF\Delta b\,\Delta D/(S-D) + O(3)$. The first-order total is $\Phi'(F_b\Delta b + F_D\Delta D)$; substitute. □

4.4 How much the split matters for a book

Proposition 4.8 (Split materiality)**.** *Let the dividend consensus error ΔL have covariance Σ_L (consensus noise) with the Totem forwards fixed. To first order the variance of the total is $g^\top\Sigma_L g$, against $s_D^\top\Sigma_L s_D$ for the dividend line alone. The share of the dividend-line uncertainty that cancels against the repo line is $1 - \sqrt{g^\top\Sigma_L g \,/\, s_D^\top\Sigma_L s_D}$.*

Proof. By Theorem 4.4 the total is the forward line, which does not involve L^{T}, plus $g \cdot \Delta L$. □

Corollary 4.9 (Offset ratio)**.** *For a forward-linear book a move of the dividend panel produces large and opposite dividend and repo lines with a small total. An offset ratio $1 - |\mathit{total}|/(|\mathit{div\ line}| + |\mathit{repo\ line}|)$ close to one therefore points to a split artefact, not to a mismark.*

g is computed from FO sensitivities and the bootstrap Jacobian only; no repricing is needed.

4.5 Worked examples

Example 4.10 (One call)**.** A one-year at-the-money call (spot 100, rate 3%, volatility 25%, FO dividend 3.5, FO repo 60 bp), with Totem forward 20 bp below FO and three different dividend panels:

Totem dividend	Implied repo (bp)	Div line	Repo line	Total	Full revaluation
3.5	80	0.000	−0.102	−0.102	−0.102
3.0	132	0.264	−0.365	−0.101	−0.102
4.2	7	−0.370	0.269	−0.101	−0.102

The implied repo moves by 125 bp across panels and each line by about 0.3, but the total does not move.

Example 4.11 (Three books)**.** The prototype pricer values three books on eleven repo pillars and six dividend periods: A (calls, puts and a forward), B (A plus a short dividend swap on 2028) and C (A plus a one-year stock-loan fee). Totem forwards are 20 bp below FO and the Totem dividend panel moves the 2027 and 2028 periods by −0.4, 0 or +0.5:

Book	Panel shift	Div line	Repo line	Total	Fwd line	Non-fwd line	Full reval.
A	−0.4	5.16	−6.91	−1.75	−1.74	−0.01	−1.74
A	0.0	0.00	−1.74	−1.74	−1.74	0.00	−1.74
A	0.5	−6.46	4.74	−1.72	−1.74	0.01	−1.74
B	−0.4	8.15	−6.91	1.24	−1.74	2.98	1.25
B	0.0	0.00	−1.74	−1.74	−1.74	0.00	−1.74
B	0.5	−10.19	4.74	−5.45	−1.74	−3.72	−5.47
C	−0.4	5.16	−5.16	0.00	−1.18	1.17	0.01
C	0.0	0.00	−1.18	−1.18	−1.18	0.00	−1.18
C	0.5	−6.46	3.82	−2.64	−1.18	−1.47	−2.66

For book A the total stays at −1.74 while each line swings by more than six units. For books B and C the total moves, by exactly the non-forward line $g \cdot \Delta L$: the dividend swap is a genuine dividend exposure and the stock-loan fee a genuine repo exposure.

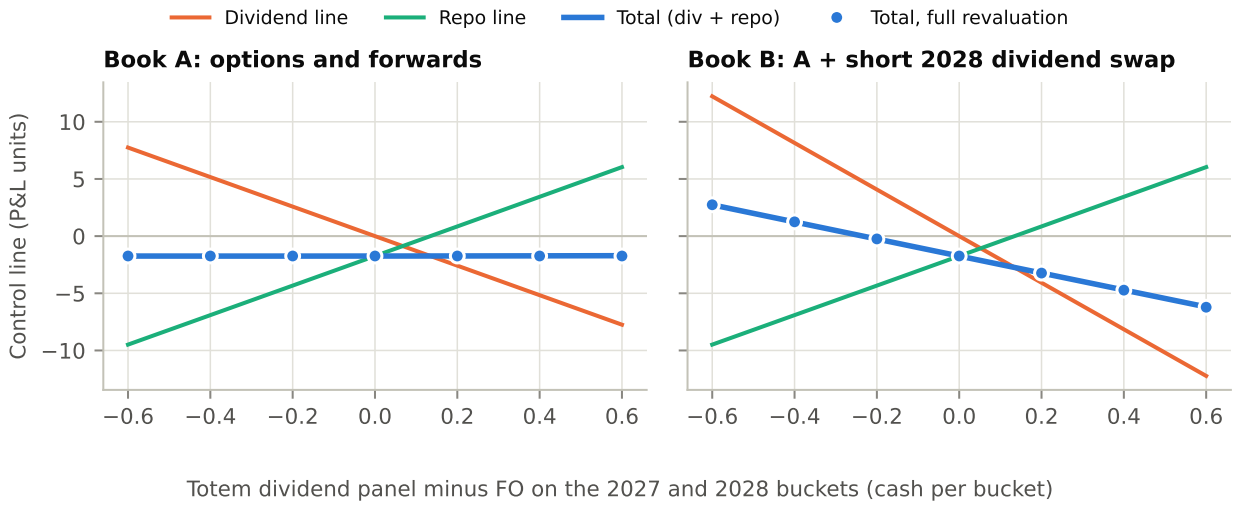


Figure 2: Dividend line, repo line and their total as the Totem dividend panel moves (book A left, book B right). Dots: full revaluation. For forward-linear exposure the total is flat (Theorems 4.4 and 4.5); the dividend swap adds a genuine dividend exposure whose line is exactly $g \cdot \Delta L$.

Example 4.12 (Materiality)**.** With a dividend consensus dispersion of 5% of each period, ten contributors and correlation $e^{-|\Delta y|/2}$ between periods, Proposition 4.8 gives

Book	sd of div line	sd of total (split risk)	Share cancelled
A	0.725	0.001	99.8%
B	1.113	0.436	60.8%
C	0.725	0.162	77.6%

99.8% of the dividend-line uncertainty cancels for book A; 60.8% and 77.6% for books B and C.

4.6 When the split does matter

The non-forward exposure g is non-zero for dividend swaps and futures (pure dividend), stock borrow and total-return-swap financing legs (pure repo), American exercise (the early-exercise boundary depends on dividends and repo separately) and exotics priced with discrete-dividend dynamics (the dividend model changes the volatility of the spot process). For those books the split must be controlled directly, and the quality of the implied repo curve (Section 6) matters.

Insight

- Control the *forward line* and the *non-forward line*: the first needs Totem forwards and FO dividends only; the second needs Totem dividends only where $g \neq 0$.
- Large offsetting dividend and repo lines with a small total are a split artefact.
- Under prudent valuation [6], if uncertainties are aggregated per valuation input without netting, a split artefact is counted twice; defining the input as the forward where $g \approx 0$ removes the double count.

5 A statistical model of the Totem–FO carry difference

5.1 Observation model

Definition 5.1 (Observation model)**.** The detector observes the carry difference $y \in \mathbb{R}^K$ of Eq. (4) and models it as

$$y = X\beta + \sum_{t\in\mathcal{S}} a_t\, t + \varepsilon, \qquad X = [\,T,\ T\ln T\,], \qquad \varepsilon \sim \mathcal{N}(0, \kappa\Sigma), \tag{12}$$

where $X\beta$ is a smooth repo difference $\delta b(T) = \beta_0 + \beta_1 \ln T$ written in carry space (legitimate disagreement, including a genuine FO mismark), $\mathcal{S}$ is a possibly empty set of anomaly templates (Section 6), Σ is the covariance of the consensus noise built below from Totem statistics, and $\kappa > 0$ is a single scale calibrated on clean history.

β_0 is the repo difference at one year ($\ln 1 = 0$); β_1 its slope in $\ln T$. Under the null hypothesis $\mathcal{S} = \emptyset$.

5.2 Covariance of consensus means

By Lemma 2.3 and Proposition 3.1, the consensus noise in carry is $\varepsilon_k = -e^F_k - e^D_k$, with e^F_k the error of the option-panel consensus $\ln \overline{F}_k$ and $e^D_k = \delta\overline{D}_{\mathcal{B}}(T_k)/(S - D_k)$ that of the allocated dividend consensus. The two panels being independent (Assumption 2.13 with $m = 0$, the conservative case),

$$\Sigma = \Sigma^F + \Sigma^D. \tag{13}$$

Both terms are covariances of means of nested panels, so Proposition 3.4 applies: each is a contributor-level covariance divided elementwise by $\max(n_k, n_l)$. What remains is to model the contributor-level covariance from the dispersions Totem publishes.

5.3 The option panel

A contributor's log-forward error at T is $-T\,\delta b_j - \delta D_j(T)/(S - D)$. Repo views are smooth (desks mark a few pillars and interpolate), so the repo part is essentially a persistent level, with covariance $\sigma_b^2 T_k T_l$. The dividend part is cumulative: a persistent factor (a contributor who is bullish on dividends is bullish on all years) plus increments that arrive with each dividend.

Lemma 5.2 (Cumulative covariance)**.** *For a non-decreasing $s \in \mathbb{R}^K_{\geq 0}$ and $w \in [0, 1]$, the matrix $C_{kl} = w\, s_k s_l + (1 - w)\min(s_k, s_l)^2$ is positive semi-definite: it is the covariance of $Z_k = \sqrt{w}\, s_k\, \xi + \sqrt{1 - w}\, W(s_k^2)$, with $\xi \sim \mathcal{N}(0, 1)$ independent of a standard Brownian motion W. For $w = 0$ the correlation between $k < l$ is s_k/s_l (independent increments).*

Proof. $\mathrm{Cov}(Z_k, Z_l) = w\, s_k s_l + (1 - w)\,\mathrm{Cov}(W(s_k^2), W(s_l^2)) = w\, s_k s_l + (1 - w)\min(s_k^2, s_l^2)$, and a covariance matrix is positive semi-definite. With $w = 0$, $\mathrm{Cov}(Z_k, Z_l)/\sqrt{\mathrm{Var}\, Z_k\, \mathrm{Var}\, Z_l} = s_k^2/(s_k s_l) = s_k/s_l$ for $k < l$. □

Proposition 5.3 (Option-panel covariance)**.** *Let $s_k = \sigma_{F,k}/F_k$ be the published dispersion of contributor forwards in log terms, and σ_b the dispersion of repo views. Splitting $s_k^2 = \sigma_b^2 T_k^2 + s_k'^2$,*

$$\Sigma^F_{kl} = \frac{\sigma_b^2\, T_k T_l + w\, s'_k s'_l + (1 - w)\min(s'_k, s'_l)^2}{\max(n_{A,k}, n_{A,l})}. \tag{14}$$

Proof. Independence of repo and dividend views gives the sum of a rank-one term and the cumulative covariance of Lemma 5.2 for the dividend part; Proposition 3.4 gives the division by the larger panel. □

Remark 5.4 (Estimating σ_b). At the shortest expiries dividend uncertainty is negligible, so $s_k \approx \sigma_b T_k$. The prototype takes the median of s_k/T_k over the first three pillars. The minimum would be biased low (the minimum of three noisy estimates), which understates the uncertainty of the repo level and makes the FO-mismark test of Section 6 fire too often; this was observed on hard-to-borrow names and corrected.

5.4 The dividend panel

Proposition 5.5 (Dividend-panel covariance)**.** *Let the period consensus errors have covariance* $(\Sigma_L)_{yz} = \rho_{yz}\,\sigma_{D,y}\sigma_{D,z}/\max(n_{B,y}, n_{B,z})$*, with* $\rho_{yz} = e^{-|y-z|/\ell}$*. Then, to first order,*

$$\Sigma^D = J_A\,\Sigma_L\,J_A^\top, \qquad (J_A)_{ky} = \frac{A_{ky}}{S - D_k}. \tag{15}$$

Proof. $e_k^D = \delta D_k/(S - D_k) = \sum_y A_{ky}\delta L_y/(S - D_k)$ by Lemma 2.11: a linear map of the period errors. □

5.5 Dispersion spikes are symptoms, not noise

Proposition 5.6 (Masking)**.** *In the timing split of Proposition 3.6, let* s_k *be the dispersion without the split and* $\Delta = \tilde{d}_i/(S - D_k)$*. The test statistic for the shift at* T_k*, computed with the published (inflated) dispersion, is*

$$z = \frac{p\,\Delta}{\sqrt{s_k^2 + p(1-p)\Delta^2}/\sqrt{n_{A,k}}}, \tag{16}$$

which stays bounded by $\sqrt{n_{A,k}\,p/(1-p)}$ *however large the dividend, whereas with the clean dispersion* z *grows linearly in* Δ*.*

Proof. Proposition 3.6(i) gives the mean shift and (iii) the inflated variance. For $\Delta \to \infty$ the ratio tends to $\sqrt{n_{A,k}}\,p/\sqrt{p(1-p)}$. □

Example 5.7. For the gallery example ($p = 0.37$, $n_A = 16$) the bound is $\sqrt{16 \times 0.37/0.63} \approx 3.1$: just around the family-wise critical value, so the split would often go unnoticed. The prototype therefore caps an isolated dispersion spike at 1.5 times the larger of its two neighbours before building Σ, and reports the ratio, which is itself evidence that contributors disagree, usually about an ex-date (here 9.7).

Insight

Σ uses only what Totem publishes (dispersions and contributor counts) plus three structural settings: the factor weight w, the correlation length ℓ of dividend periods and the repo dispersion read off the short end. The single scale κ absorbs a common bias of these settings; their *shape* is what makes the detector powerful, because it tells it which deviations are cheap (smooth, long-dated) and which are expensive (short-dated, local).

6 Detection: matched filters and the likelihood-ratio test

6.1 Templates

Each mechanism of Section 3 leaves a known shape in carry space. The detector tests one *template* $t \in \mathbb{R}^K$ per mechanism and location:

Template	Vector	Mechanisms	Amplitude a in natural units
level	$\mathbf{1}$	spot snapshot; dividend before T_1	spot error $\approx -a$; front dividend $\approx -a(S - D_1)$
step k	$\mathbf{1}\{T \geq T_k\}$	dividend amount in $(T_{k-1}, T_k]$; panel change at T_k	dividend error $\approx -a(S - D_k)$
spike k	e_k	forward outlier; ex-date timing across T_k	$\delta \ln F_k \approx -a$; timing share $\approx -a(S - D_k)/\tilde{d}_i$
season	$\sum_i \tilde{d}_i K_h(T_k - t_i)/(S - D_k)$	ex-date profile shift	shift $\approx 365\, a$ days

K_h is a Gaussian kernel ($h = 20$ days). The natural units follow from Propositions 3.6 and 3.8 to 3.10. A spike at the last pillar is the same vector as a step there and is reported as either. With eleven pillars there are 22 templates.

6.2 Whitening, projection and the matched filter

Let $\kappa\Sigma = LL^\top$ (Cholesky), $\tilde{y} = L^{-1}y$, $\tilde{X} = L^{-1}X$, and $P = I - QQ^\top$ with Q an orthonormal basis of the columns of $\tilde{X}$: P removes the null model in whitened space. Write $r = P\tilde{y}$ and $\tilde{t} = PL^{-1}t$.

Theorem 6.1 (Generalised likelihood-ratio test for one template)**.** *In the model $y = X\beta + a\,t + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \kappa\Sigma)$, with β unknown, the generalised likelihood-ratio statistic for $H_0 : a = 0$ against $H_1 : a \neq 0$ is z_t^2 with*

$$z_t = \frac{\tilde{t}^\top r}{\|\tilde{t}\|}, \qquad z_t \sim \mathcal{N}(a\|\tilde{t}\|,\, 1). \tag{17}$$

The maximum-likelihood amplitude is $\hat{a}_t = \tilde{t}^\top r/\|\tilde{t}\|^2$ with standard error $1/\|\tilde{t}\|$. Among tests invariant to the null-model subspace and to the sign of a, the test that rejects for large $|z_t|$ is uniformly most powerful.

Proof. After whitening, $\tilde{y} = \tilde{X}\beta + a\,L^{-1}t + \tilde{\varepsilon}$ with $\tilde{\varepsilon} \sim \mathcal{N}(0, I)$. The maximised log-likelihood is $-\frac{1}{2}$ times the residual sum of squares, which is $\|P\tilde{y}\|^2$ under H_0 and the residual of the regression on $[\tilde{X}, L^{-1}t]$ under H_1. By the Frisch–Waugh–Lovell theorem [4] the latter equals $\|r\|^2 - (\tilde{t}^\top r)^2/\|\tilde{t}\|^2$, so $2\ln\Lambda = (\tilde{t}^\top r)^2/\|\tilde{t}\|^2 = z_t^2$, and $\hat{a}_t$ is the coefficient of the partialled-out regression. Since $P\tilde{X} = 0$ and $\tilde{t}^\top P = \tilde{t}^\top$, $\tilde{t}^\top r = a\,\tilde{t}^\top L^{-1}t + \tilde{t}^\top\tilde{\varepsilon} = a\|\tilde{t}\|^2 + \tilde{t}^\top\tilde{\varepsilon}$, and $\tilde{t}^\top\tilde{\varepsilon}/\|\tilde{t}\| \sim \mathcal{N}(0,1)$. Optimality is the matched-subspace detector result of Scharf and Friedlander [7]; see also [3]. □

Corollary 6.2 (Power and signal-to-noise)**.** *At threshold c the detection probability is $\Phi(\lambda - c) + \Phi(-\lambda - c)$, where $\lambda = |a|\,\|\tilde{t}\|$ is the signal-to-noise ratio of the anomaly. With the typical family-wise threshold $c \approx 3.39$ of the benchmark, an anomaly with $\lambda = 4$ is detected with probability 73.0 % and one with $\lambda = 6$ with 99.5 %.*

Proof. $|z_t| > c$ with $z_t \sim \mathcal{N}(\lambda, 1)$ (up to sign). □

Proposition 6.3 (Global test)**.** *Under H_0, $Q = \|r\|^2 \sim \chi^2_{K-p}$ with $p = 2$ the number of null-model columns.*

Proof. $r = P\tilde{\varepsilon}$ with P an orthogonal projection of rank $K - p$. □

Q catches roughness that no template explains.

6.3 Family-wise calibration

Proposition 6.4 (Family-wise critical value)**.** *Under H_0 the vector $z = (z_t)_t$ is Gaussian with mean zero and correlation $C_{tu} = \tilde{t}^\top \tilde{u}/(\|\tilde{t}\|\|\tilde{u}\|)$, which depends neither on β nor on κ. The critical value c_α with $P(\max_t |z_t| > c_\alpha) = \alpha/2$ is computed exactly by simulating $\mathcal{N}(0, C)$, and is at most the Šidák value $\Phi^{-1}(1 - \frac{1}{2}[1 - (1 - \alpha/2)^{1/m}])$ for m templates. The rule "alarm if $\max_t |z_t| > c_\alpha$ or Q exceeds its $1 - \alpha/2$ quantile" has false-alarm rate at most α.*

Proof. $z = D^{-1}\tilde{T}^\top P\tilde{\varepsilon}$ with $\tilde{T}$ the matrix of the $\tilde{t}$ and D their norms, hence the correlation. Šidák's inequality [8] gives $P(\max_t |z_t| \le c) \ge \prod_t P(|z_t| \le c)$ for any correlation, so the Šidák threshold is conservative. The two tests share the budget α by the union bound. □

Proposition 6.5 (Exact calibration of the variance scale)**.** *Replacing $\kappa = 1$ by κ divides every z_t by $\sqrt{\kappa}$ and Q by κ, and leaves c_α unchanged. For a set of clean names the alarm indicator is therefore non-increasing in κ, the empirical alarm rate is a non-increasing step function of κ, and the smallest κ whose alarm rate does not exceed the target is found exactly by bisection on $\ln \kappa$.*

Proof. Whitening with $\sqrt{\kappa}L$ scales $\tilde{y}$, $\tilde{X}$ and every $\tilde{t}$ by $\kappa^{-1/2}$; P is unchanged, so r and $\tilde{t}$ scale by $\kappa^{-1/2}$, z_t by $\kappa^{-1/2}$ and Q by κ^{-1}; C is a correlation and does not move. Each name's indicator $\{\max |z|/\sqrt{\kappa} > c\} \cup \{Q/\kappa > q\}$ shrinks as κ grows. □

6.4 Attribution, repair and the FO-mismark test

After an alarm, templates are selected greedily (matching pursuit [5]): at each step the template with the largest $|z|$ after projecting out the null model and the templates already chosen is added, while it exceeds c_α. The selected templates are then refitted jointly with the null model by generalised least squares, each amplitude is translated into natural units, and the *repaired* carry is $y - \sum_{t \in \mathcal{S}} \hat{a}_t\, t$. Templates inside the span of the null model are undetectable by design: a smooth repo difference is legitimate disagreement, reported but never flagged.

The repo level $\hat{\beta}_0$ (Totem minus FO at one year) and its standard error come from the same fit. The FO repo is reported as a candidate mismark when $|\hat{\beta}_0|$ exceeds both 10 bp and $t_{\nu,1-\alpha/2} \cdot s \cdot \mathrm{se}(\hat{\beta}_0)$:

- t_ν is a Student quantile with $\nu = n_A - 1$ at the short end: the standard error rests on dispersions estimated from a handful of contributors, and a Gaussian estimate divided by an independent χ^2_ν-based scale is Student distributed;
- $s \ge 1$ is a scale calibrated on clean history exactly like κ, so that clean names raise at most α FO-mismark flags (1.00, 1.05, 1.00 and 1.18 for the four categories of the benchmark).

Example 6.6 (A detection by hand)**.** Four pillars $T = (0.25, 0.5, 1, 2)$, independent noise with standard deviations $(1, 2, 4, 8)$ bp of carry, a null model with the repo level only ($X = T$), a true level of 30 bp, a spike of 12 bp at T_2 and noise $(0.4, -1.1, 2.6, -3.0)$ bp, so $y = (7.9, 25.9, 32.6, 57.0)$ bp. The fitted level is $\hat{\beta}_0 = 36.1$ bp and $Q = 21.0$ (the 99% quantile of χ^2_3 is 11.3). The 7 templates give

Template t	z_t	$\hat{a}_t$ (bp)	s.e. (bp)
Level	0.55	0.82	1.49
Step from T_2	4.06	10.98	2.70
Step from T_3	−2.21	−10.67	4.82
Step at T_4	−2.20	−20.33	9.24
Spike at T_1	−1.31	−1.51	1.15
Spike at T_2	4.52	10.45	2.31
Spike at T_3	−1.02	−4.70	4.62

The family-wise critical value at $\alpha = 2\%$ is $c = 3.11$ (Šidák: 3.19). The spike at T_2 wins ($z = 4.52$, amplitude 10.5 ± 2.3 bp against 12 injected); the step from T_2 also scores high because the two templates are correlated. Matching pursuit adds the spike, after which every remaining $|z|$ is below 0.8 and Q falls to 0.6: one explanation suffices.

6.5 Algorithms

Algorithm 1 Implied repo bootstrap

Require: spot S, cumulative rate $R(\cdot)$, dividends (t_i, d_i), pillars $T_1 < \cdots < T_K$, forwards $F_1, \ldots, F_K$

for $k = 1, \ldots, K$ **do**
  build Φ_k of Eq. (3) from $B_1, \ldots, B_{k-1}$
  $x \leftarrow R_k - \ln\big(F_k/(S - \hat{D}_k)\big)$ ▷ in-period dividends at the previous repo
  repeat
    $g \leftarrow \ln \Phi_k(x) - \ln F_k$; update the bracket $[\ell, u]$ with the sign of g
    $x \leftarrow x - g\,\Phi_k(x)/\Phi_k'(x)$; if $x \notin (\ell, u)$, $x \leftarrow (\ell + u)/2$
  until the step is below tolerance
  $B_k \leftarrow x$
return $b_k = B_k/T_k$

Algorithm 2 Structural detection

Require: carry difference y, covariance Σ, scale κ, null design X, templates $\mathcal{T}$, level α

$L \leftarrow \mathrm{chol}(\kappa\Sigma)$; $P \leftarrow$ projector removing $L^{-1}X$; $r \leftarrow PL^{-1}y$; $Q \leftarrow \|r\|^2$
for each $t \in \mathcal{T}$: $\tilde{t} \leftarrow PL^{-1}t$, $z_t \leftarrow \tilde{t}^\top r/\|\tilde{t}\|$
$c_\alpha \leftarrow$ the $1 - \alpha/2$ quantile of $\max_t |Z_t|$, $Z \sim \mathcal{N}(0, C)$ ▷ Proposition 6.4
alarm $\leftarrow \max_t |z_t| > c_\alpha$ **or** $Q > \chi^2_{K-2}(1 - \alpha/2)$
if alarm **then**
  $\mathcal{S} \leftarrow \emptyset$; **repeat** add the template with the largest $|z|$ after projecting out $[X, \mathcal{S}]$ **while** that $|z|$ exceeds c_α
joint GLS fit of y on $[X, \mathcal{S}]$; translate amplitudes into natural units; repaired carry $y - \sum \hat{a}_t t$
FO-mismark flag if $|\hat{\beta}_0| > \max\big(10\,\mathrm{bp},\ t_\nu\, s\,\mathrm{se}(\hat{\beta}_0)\big)$

Algorithm 3 Calibration of the variance scale

Require: clean names $i = 1, \ldots, N$ with $(\max_t |z_{t,i}|,\ c_i,\ Q_i)$ computed at $\kappa = 1$, target α

$\ell \leftarrow \ln 10^{-3}$, $u \leftarrow \ln 10^3$
for 60 iterations **do**
  $m \leftarrow (\ell + u)/2$; rate $\leftarrow \frac{1}{N}\sum_i \mathbf{1}\{\max|z_i|e^{-m/2} > c_i$ or $Q_i e^{-m} > q_{1-\alpha/2}\}$
  if rate $\le \alpha$ **then** $u \leftarrow m$ **else** $\ell \leftarrow m$
return $\kappa = e^u$

6.6 Examples on synthetic names

Figure 3 shows the machinery in carry space for a dividend bucket error, and Fig. 4 six detections in repo space with the detector's explanations in Table 1.

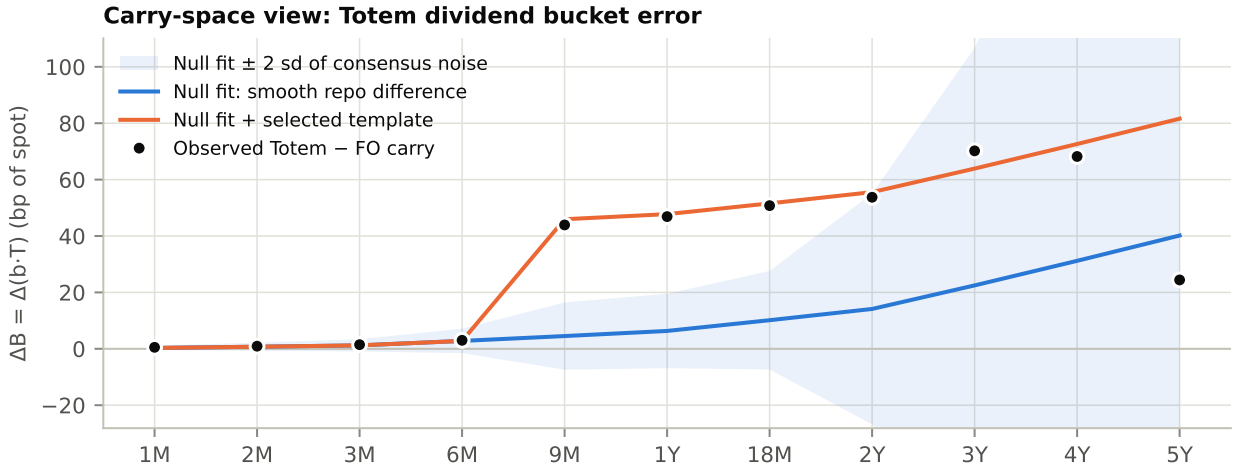


Figure 3: Carry-space view of the Totem dividend bucket error of Fig. 4(b): observed carry difference, null fit with its ±2 standard-deviation band, and null fit plus the selected step template.

Table 1: First explanation returned by the detector for each panel of Fig. 4.

Panel	First explanation of the structural detector
(a)	Level shift in carry (1/T in repo): reference spot vs FO spot off by -0.10% or a dividend before 1M off by -0.103
(b)	Step from 9M: dividends ex in (6M, 9M] look -0.400 (-11%) off in the Totem dividend versus the forwards
(c)	Dividend timing at 6M: about 37% of the 2.18 dividend ex +17d from expiry is on the other side of the expiry (forward dispersion at 6M is 9.7x its neighbours: contributors disagree)
(d)	Ex-date profile shift of about -55 days between the dividend allocation and the forwards
(e)	Forward outlier at 3Y: Totem ln F off by +129bp
(f)	No Totem anomaly. FO repo level differs from Totem by +16bp (± 6bp): FO mismark candidate

6.7 Small neural networks are matched filters

Proposition 6.7. *Let the null model be projected out and the noise whitened, so that* $r \sim \mathcal{N}(0, P)$ *under* H_0 *and* $r \sim \mathcal{N}(a\tilde{t}, P)$ *under* H_1 *with a known template* t *and amplitude* a*. The Bayes posterior log-odds is*

$$\ln \frac{P(H_1 \mid r)}{P(H_0 \mid r)} = a\,\tilde{t}^{\top} r - \tfrac{1}{2} a^2 \|\tilde{t}\|^2 + \ln \frac{\pi_1}{\pi_0}, \tag{18}$$

a linear function of the matched filter $\tilde{t}^{\top} r$*. A classifier trained by cross-entropy on enough data converges to this posterior, so in this setting a "small network per anomaly type" can at best learn the matched filter; with an unknown amplitude and sign the likelihood-ratio statistic* $|z_t|$ *of Theorem 6.1 emerges.*

Proof. The two Gaussian densities share the covariance P, so their log-ratio is linear in r; the quadratic terms cancel. Cross-entropy is minimised by the true posterior. Maximising Eq. (18) over a gives $(\tilde{t}^{\top} r)^2/(2\|\tilde{t}\|^2)$. □

Insight

The detector is the optimal one for the mechanisms we can name, written down instead of learned: its false-alarm rate is set exactly, its power is known in closed form (Corollary 6.2), and every alarm comes with an amplitude in natural units. Learned models earn their place on what the templates do not cover (Section 8).

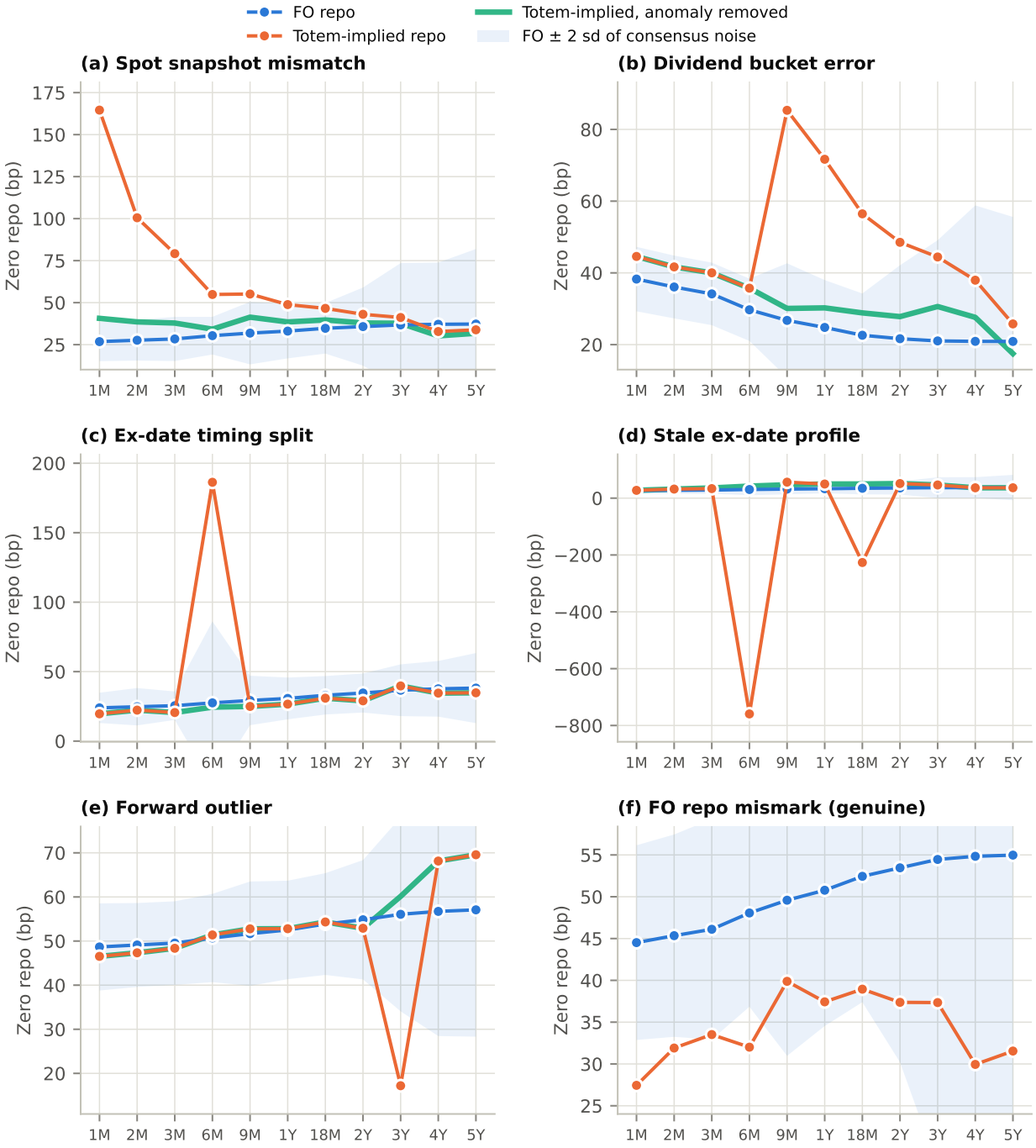


Figure 4: Six synthetic EU single stocks, one per scenario, with the benchmark's detector settings: FO repo, Totem-implied repo, the implied repo with the detected anomaly removed, and the per-pillar noise band. In (f) the Totem curve is fine and the FO repo is shifted: no anomaly is raised and the FO mismark is reported; the band is per pillar while the mismark test pools all pillars.

7 Benchmark on a synthetic Totem market

7.1 Design

The synthetic market is built from contributors, so that every mechanism of Section 3 arises the way it does in Totem rather than as a shape added to a curve.

Categories. Four categories with eleven Totem expiries from one month to five years and six dividend periods (the rest of 2026, then 2027 to 2031); valuation date 14 October 2026.

Category	Dividends	Yield	Repo (bp), short → long	Option panel	Div. panel
EU stock	Annual, Apr–May (35% with a Nov. interim)	2.0%–6.0%	$[0, 50] \to [0, 60]$	$16 \to 6$	$12 \to 4$
US stock	Quarterly	0.5%–3.0%	$[0, 40] \to [0, 50]$	$18 \to 5$	$10 \to 3$
Hard to borrow	Annual or quarterly	0.0%–2.0%	$[150, 800] \to [30, 150]$	$12 \to 4$	$8 \to 3$
EU index	Monthly stream, 60% in Apr–Jun	2.8%–3.6%	$[-20, 60] \to [-10, 80]$	$25 \to 12$	$20 \to 10$

Contributors. The option and dividend panels are drawn independently. Each contributor holds internally consistent views (Assumption 2.12): dividends with a persistent factor plus per-dividend noise, with a dispersion that grows with time, and a smooth repo curve (level plus tilt). Option contributors price calls and puts at five strikes with their own forward, a skewed volatility surface, volatility noise and price noise. Panels shrink with tenor and are nested.

Consensus and pipeline. The consensus is the mean after a median/MAD screen; dispersions and counts are reported as Totem would. The controller pipeline then runs unchanged: parity forwards, allocation on the FO ex-date profile, bootstrap, covariance, detection.

FO marks. True dividends with 1% noise and the true repo with a 3 bp level error (one standard deviation).

Scenarios. Each test name is simulated under every scenario with the same seed, so differences between scenarios are paired.

Scenario	Injection	Magnitude
Spot snapshot mismatch	option contributors price off a different spot	±0.03% to ±0.15%
Dividend bucket error	one period (2027–2029) off in the dividend consensus	±0.15 to ±0.5 per 100 of spot
Ex-date timing split	35–70% of the option panel moves the dividend closest to an expiry to its other side	one dividend
Stale ex-date profile	allocation profile shifted so that one dividend crosses an expiry	3–20 days past it
Forward outlier	consensus calls at one expiry shifted	3–6 s.d. of the carry noise
Biased long-dated sub-panel	contributors who remain after the largest panel drop are biased on later dividends	±8% to ±20%
FO repo mismark	FO repo shifted in parallel, Totem untouched	±15 to ±50 bp
Gross/net bias	every Totem period scaled	±8% to ±18%

Detectors. (1) Naive gap: largest $|\Delta b|$ over the pillars. (2) Naive shape: largest deviation of the Totem-implied repo from the interpolation of its two neighbours. (3) B-spline plus Gaussian mixture on the Totem curve: least-squares cubic B-spline coefficients (six, knots in $\ln T$) clustered by a Gaussian mixture per category with the number of components chosen by BIC; the score is the negative log-likelihood. (4) The same on the Totem–FO difference. (5) The structural detector of Section 6.

Protocol. 300 clean names per category calibrate every detector to 2% false alarms (a quantile of the score for the baselines, κ and the level scale for the structural detector). 150 fresh names per category are then simulated under each of the 9 scenarios, 5400 test names in total. Because the design is paired, the signal-to-noise of an injection is computed exactly as $\lambda = \|PL^{-1}(y^{\text{scenario}} - y^{\text{clean}})\|$, the quantity of Corollary 6.2.

7.2 Results

Table 2: Share of names flagged, every detector calibrated to 2% false alarms on clean history. The clean row is the realised false-alarm rate on fresh names. For the FO mismark and the gross/net bias a flag is not the desired outcome (see text). Columns: naive gap, naive shape, B-spline plus Gaussian mixture on the curve and on the Totem–FO difference, structural detector.

Scenario	Naive gap	Naive shape	GMM curve	GMM diff.	Structural
Clean (null)	2%	3%	2%	3%	3%
Spot snapshot mismatch	89%	64%	96%	100%	98%
Totem dividend bucket error	29%	41%	47%	56%	55%
Ex-date timing split in option panel	55%	74%	65%	69%	71%
Stale ex-date profile in allocation	90%	92%	94%	94%	94%
Forward outlier at one expiry	35%	87%	74%	78%	97%
Biased long-dated sub-panel	42%	24%	36%	39%	24%
FO repo mismark (genuine)	60%	3%	2%	75%	3%
Gross/net dividend bias (all buckets)	70%	71%	80%	86%	88%
Mean over the six Totem anomalies	**57%**	**64%**	**69%**	**73%**	**73%**

Table 3: Area under the ROC curve against the clean names (ranking quality, independent of the threshold).

Scenario	Naive gap	Naive shape	GMM curve	GMM diff.	Structural
Spot snapshot mismatch	0.98	0.85	1.00	1.00	1.00
Totem dividend bucket error	0.70	0.72	0.81	0.86	0.84
Ex-date timing split in option panel	0.77	0.87	0.84	0.87	0.89
Stale ex-date profile in allocation	0.96	0.96	0.97	0.97	0.97
Forward outlier at one expiry	0.71	0.83	0.87	0.90	0.99
Biased long-dated sub-panel	0.75	0.63	0.76	0.80	0.74
FO repo mismark (genuine)	0.85	0.50	0.50	0.96	0.50
Gross/net dividend bias (all buckets)	0.90	0.87	0.97	0.98	0.97

7.3 Reading the results

- **Average detection does not separate the best methods.** Over the six Totem-side anomalies the structural detector flags 73% of names, B-spline plus GMM on the difference

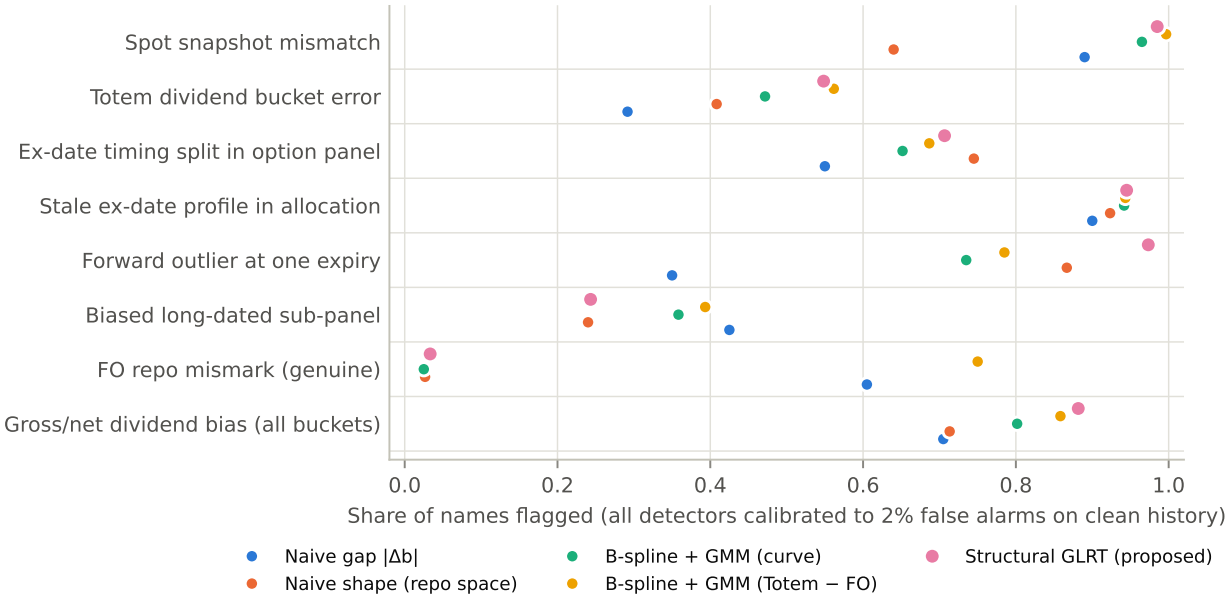


Figure 5: Share of names flagged per scenario and detector (the data of Table 2).

Table 4: Structural detector: share of names flagged per scenario and category.

Scenario	EU stock	US stock	Hard to borrow	EU index
Clean (null)	3%	5%	3%	1%
Spot snapshot mismatch	100%	100%	94%	100%
Totem dividend bucket error	51%	65%	27%	76%
Ex-date timing split in option panel	81%	90%	57%	55%
Stale ex-date profile in allocation	100%	100%	95%	83%
Forward outlier at one expiry	99%	99%	100%	91%
Biased long-dated sub-panel	12%	17%	5%	63%
FO repo mismark (genuine)	4%	5%	3%	1%
Gross/net dividend bias (all buckets)	100%	100%	53%	100%

73%, on the curve 69%, the naive shape check 64% and the naive gap 57%.

- **Forward outliers** are where carry space and the noise model pay most: 97% detected against 35% for the naive gap, which cannot see a three-to-six standard deviation outlier at a long expiry under the long-end noise, and 87% for the naive shape check.
- **Spot mismatches** (98%) and **stale ex-date profiles** (94%) are easy for every method; the structural detector adds the explanation (Fig. 7).
- **Dividend bucket errors** (55% against 29% naive) are limited by their size: an error of 0.15 to 0.5 per 100 of spot is small next to long-dated dividend dispersion, and Fig. 6 shows that the misses are low signal-to-noise cases.
- **Ex-date timing splits** are detected in 71% of names; the naive shape check (74%) is competitive because these spikes are large in repo space.
- **A biased long-dated sub-panel** is close to non-identifiable for every method (24% structural, 42% naive gap): the bias builds up smoothly after the panel drop and mostly projects onto a repo slope. This is Theorem 3.12 at work, not a tuning issue.
- **Genuine FO mismarks** are kept apart from bad Totem curves: 3% raise an anomaly (the clean rate) and 80% are reported as FO-mismark candidates, against 2% of clean names. The naive gap and GMM on the difference raise a bad-curve alarm on 60% and 75% of them.

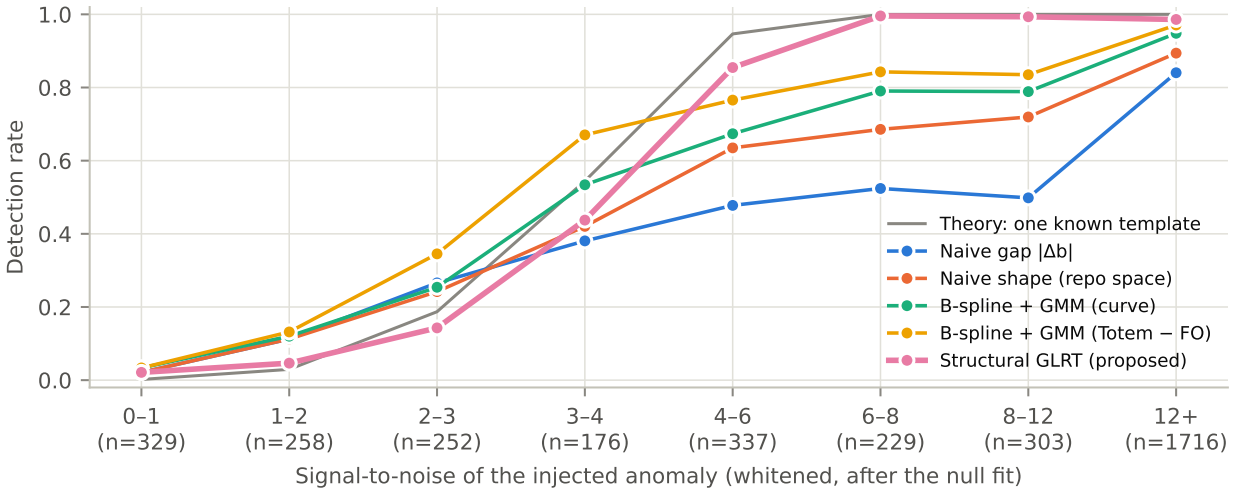


Figure 6: Detection rate against the signal-to-noise ratio λ of the injected anomaly, pooled over the six Totem-side anomalies; n is the number of names per bin. The grey line is the theoretical power of a single known template at the typical family-wise threshold (Corollary 6.2).

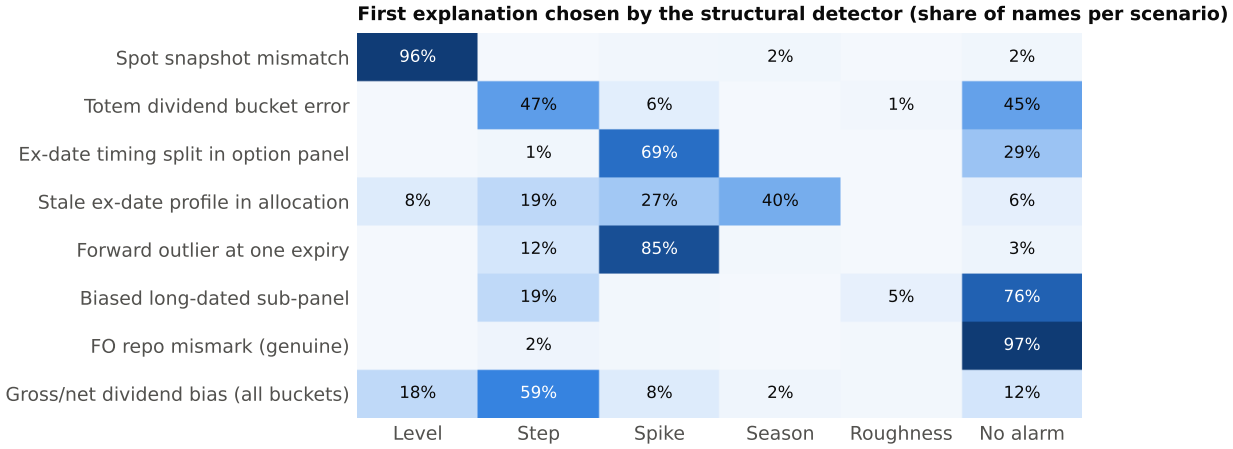


Figure 7: Mechanism named first by the structural detector, per scenario. Level, step, spike and season are the template families; roughness means the global test fired without a template; no alarm means the name was not flagged.

The FO-mismark test finds 97% of the mismarks on EU stocks and 100% on the index, but only 24% on hard-to-borrow names: there contributors' repo views differ by about 60 bp, so a 15–50 bp FO difference is within the uncertainty of the consensus itself.

- **Gross/net bias** is flagged in 88% of names, mostly as steps at ex-dates for discrete payers; for an index dividend stream it looks like a repo slope and is reported through β instead (Proposition 3.11).
- **Power.** Between signal-to-noise 4 and 6 the structural detector finds 85% of anomalies, above 6 100% to 99%, close to the theory of Corollary 6.2. Below 3 the naive gap and GMM on the difference detect more (35% against 14% between 2 and 3): they also react to the part of an anomaly that looks like a smooth repo difference, which the structural detector leaves to the FO-mismark test on purpose. That same property is why they raise alarms on genuine FO mismarks.
- **Attribution.** When the structural detector flags a Totem anomaly, its first explanation names the injected mechanism and location in 88% of cases.

Table 5: Structural detector: detection, attribution (the first explanation names the injected mechanism and its location) and FO-mismark flags.

Scenario	Flagged	Right mechanism (of flagged)	FO-mismark flag
Spot snapshot mismatch	98%	97%	2%
Totem dividend bucket error	55%	80%	7%
Ex-date timing split in option panel	71%	97%	2%
Stale ex-date profile in allocation	94%	71%	23%
Forward outlier at one expiry	97%	97%	2%
Biased long-dated sub-panel	24%	66%	2%
FO repo mismark (genuine)	3%	–	80%
Gross/net dividend bias (all buckets)	88%	–	41%

Table 6: Realised false-alarm rate on fresh clean names, by category (target 2%), and the calibrated κ.

Category	Naive gap	Naive shape	GMM curve	GMM diff.	Struc-tural	κ
EU stock	4.0%	3.3%	1.3%	4.0%	3.3%	1.60
US stock	3.3%	4.0%	1.3%	1.3%	5.3%	1.88
Hard to borrow	1.3%	1.3%	1.3%	4.0%	3.3%	0.96
EU index	1.3%	2.0%	6.0%	2.0%	0.7%	3.58

7.4 Limits of the evidence

Caveat

The synthetic market reproduces the mechanisms by construction; it does not show that real Totem data contains only these. Real data will also carry stale contributors, corporate actions and special dividends, American exercise in single-stock quotes and data-entry errors. The scale κ absorbs a misspecified noise level but not a misspecified structure, and the magnitudes, while plausible, are not calibrated to any particular history. The benchmark shows that the method works when the mechanisms are present and is calibrated when they are not; validation on twelve to twenty-four months of real snapshots with labelled incidents is the next step.

8 Where B-splines, Gaussian mixtures and neural networks fit

8.1 What B-spline coefficients cannot see

Proposition 8.1 (Leverage of a spike)**.** *Let $B \in \mathbb{R}^{K\times q}$ be the B-spline design matrix at the pillars and $H = B(B^\top B)^{-1}B^\top$ the hat matrix. The least-squares fit of a unit spike e_k has energy $\|He_k\|^2 = h_{kk}$, and $\sum_k h_{kk} = q$. A fraction $1 - h_{kk}$ of the spike's energy stays in the residual and is invisible to any feature computed from the coefficients.*

Proof. H is symmetric and idempotent, so $\|He_k\|^2 = e_k^\top H^\top He_k = e_k^\top He_k = h_{kk}$, and $\sum_k h_{kk} = \operatorname{tr} H = \operatorname{rank} B = q$. The coefficients $(B^\top B)^{-1}B^\top e_k$ determine He_k and nothing else. □

Example 8.2. For the six cubic B-splines (knots in $\ln T$) of the benchmark and the eleven pillars,

Pillar	1M	2M	3M	6M	9M	1Y	18M	2Y	3Y	4Y	5Y
h_{kk}	1.00	0.80	0.48	0.58	0.32	0.38	0.37	0.34	0.51	0.37	0.86
$1 - h_{kk}$	0.00	0.20	0.52	0.42	0.68	0.62	0.63	0.66	0.49	0.63	0.14

At the interior pillars between nine months and four years the coefficients capture only 0.32 to 0.80 of a spike's energy (on average $6/11 \approx 0.545$ over all pillars). The spline also spreads what it captures over several coefficients. A one-pillar anomaly, the most common bad-curve pattern, is therefore diluted before any clustering starts.

8.2 Why a mixture on the difference flags FO mismarks

Proposition 8.3. *Let the features be the spline coefficients $c = B^{+}\Delta b$ of the repo difference. A parallel FO repo shift δ moves them by $-\delta\, B^{+}\mathbf{1}$, a fixed direction. If the mixture is fitted on clean history, where FO and Totem agree up to noise, its density decreases along that direction, so its alarm rate on FO mismarks tends to one as $|\delta|$ grows, whatever the quality of the Totem curve.*

Proof. c is linear in Δb, and Δb moves by $-\delta\mathbf{1}$. Each Gaussian component has a finite variance along $B^{+}\mathbf{1}$, so the mixture density, and hence the likelihood score, goes to zero along that direction. □

The structural detector has no such problem: $T \cdot \mathbf{1}$ lies in the span of the null model, so a level difference is projected out of every test and reported through $\hat{\beta}_0$. The benchmark confirms both statements: 75% of FO mismarks are flagged by the mixture on the difference, 3% by the structural detector.

8.3 Where they help

- **P-splines in the null model.** If $\beta_0 + \beta_1 \ln T$ proves too rigid on real data, a penalised spline [1] can replace it as the smooth part of the null model; the tests of Section 6 are unchanged.
- **Mixtures on structured outputs.** A Gaussian mixture on the detector's outputs ($\hat{\beta}_0$, $\hat{\beta}_1$, the z_t, Q and the dispersion ratios) learns the usual regimes per region and category in a low-dimensional space where each coordinate means something.
- **Region × category** is best used to group the calibration of κ and of the level scale, not as clusters.
- **Neural networks.** By Proposition 6.7, on the mechanisms we can name a network can at best learn the matched filter. Networks earn their place once there is a labelled incident history and patterns the templates do not cover; they should then be trained on residuals of the structural model and on synthetic injections, not on raw curves.

9 Implementation and operations

9.1 Package

The Python package `repo_control` (numpy and scipy; scikit-learn only for the mixture baseline) implements every object of this note:

Module	Content
`curves`	zero curves with flat forwards, discrete dividends, forwards, the bootstrap of Algorithm 1
`parity`	forward and discount factor from consensus call and put prices (Lemma 2.7)
`buckets`	dividend periods, allocation, allocation matrix A (Lemma 2.11)
`noise`	panel gap, nested-panel covariance, option- and dividend-panel covariance, dispersion capping (Section 5)
`pipeline`	Totem and FO snapshots $\to$ implied repo, carry difference, Σ, detection context
`detect`	templates, matched filters, family-wise and κ calibration, attribution, repair, FO-mismark test
`control`	control lines, bootstrap Jacobian J, split materiality (Section 4)
`pricer`	a minimal pricer producing bucketed sensitivities for the examples
`synthetic`, `benchmark`, `baselines`	contributor-level synthetic market and the benchmark of Section 7

Every component has a closed-form check in the test suite: bootstrap round trip, parity, the control identity against full revaluation, the panel-gap and drop-off formulas, null calibration of the detector and recovery of each template.

9.2 Production inputs

Input	Fields	Shape
Totem options	expiries; strikes; consensus call and put prices; contributors; dispersion of contributor forwards (or an estimate); reference spot	$K \times M$
Totem dividends	period edges; consensus level; dispersion; contributors (net of dividends already paid)	Y
FO market	spot at the Totem snapshot time; rate curve; repo zero curve at the Totem expiries; dividend forecast	K; list
FO risk	repo sensitivities by pillar (mapped to the Totem expiries); dividend sensitivities by Totem period	$K+Y$ per book
History	twelve to twenty-four monthly snapshots and known incidents	for κ and validation

If the dispersion of contributor forwards is not published, it can be estimated from the dispersion of deep in-the-money calls and puts or calibrated on history; the detector needs the right scale per expiry, and κ corrects a common bias.

9.3 Monthly procedure

1. Load the Totem and FO snapshots at the same valuation time; check the reference spot.
2. Imply forwards by parity; review the implied discount factor, the parity residuals and the dispersion ratios.
3. Allocate the Totem dividend periods on the FO ex-date profile; imply repo with Totem dividends and with FO dividends.
4. Build Σ and run the detector with the κ and level scale of the region and category.
5. Compute the control lines per book: forward, dividend, non-forward and repo lines; check split materiality.
6. Review alarms with their explanations and FO-mismark candidates; use the repaired curve where a Totem anomaly is confirmed.

7. Record the decision for each alarm; the labels feed the next calibration.

9.4 Calibration and governance

κ and the level scale are recalibrated quarterly per region and category on clean history, with an alarm budget of 2% of names per month. The realised alarm rate and the share of explanations accepted by reviewers are monitored. Where they trade, exchange anchors complement the detector: listed index and dividend futures, total-return futures that quote a financing spread, and securities-lending fees for the short end of single stocks.

10 Conclusion

Implied repo curves are irregular because they are the residual of two consensus means computed on different panels; part of that residual cannot be removed by any method that sees only consensus data (Theorem 3.12). Two consequences follow. For forward-linear exposure the total control does not need the split, exactly on full revaluation and to first order on control lines (Theorems 4.4 and 4.5), and the non-forward exposure g tells, book by book, how much the split matters. For detection, the mechanisms that do produce bad curves have known signatures in carry space; matched filters with a noise model built from Totem's own statistics detect them at a calibrated false-alarm rate, explain them in natural units and keep genuine FO mismarks apart. The next step is to run the pipeline on twelve to twenty-four months of real Totem and FO snapshots, calibrate the settings per region and category, and measure attribution against labelled incidents.

A Notation

Symbol	Meaning
S, r, $R = rt$	spot, zero rate, cumulative rate
b, $B = bt$	zero repo, cumulative repo (carry)
$G = e^{R-B}$	carry growth factor
t_i, d_i, $\tilde{d}_i = d_i/G(t_i)$	ex-dates, cash dividends, carry-discounted dividends
$D(T)$	carry-discounted cumulative dividend, Eq. (1)
T_k, $k = 1..K$	Totem expiries (pillars)
L_y, π_i, A_{ky}	dividend level of period y, profile weights, allocation matrix
$\mathcal{A}$, $\mathcal{B}$, n_A, n_B, m	option and dividend panels, their sizes, contributors common to both
$\overline{b}_{\mathcal{A}}$, $\overline{D}_{\mathcal{A}}$, $\overline{D}_{\mathcal{B}}$	option-panel average repo and dividend views; dividend-panel consensus
y_k	carry difference Totem minus FO, Eq. (4)
$X = [T, T\ln T]$, β	null model (smooth repo difference) and its coefficients
Σ, κ	consensus-noise covariance, variance scale
t, $\tilde{t}$, z_t, Q	template, its whitened projection, matched-filter statistic, global chi-square
s_b, s_D, J, g	FO sensitivities to repo and dividends, implied-repo Jacobian, non-forward exposure

B Additional proofs and remarks

Lemma B.1 (Heterogeneous discounting)**.** *If contributor j discounts with DF_j, the zero crossing of $\overline{C} - \overline{P}$ is $K^* = \sum_j DF_j F_j / \sum_j DF_j$, a discount-weighted mean of the contributors' forwards. Its difference from the plain mean is* $\mathrm{Cov}(DF_j, F_j)/\overline{DF}$*, of second order.*

Proof. $\overline{C} - \overline{P} = \frac{1}{n}\sum_j DF_j(F_j - K)$ vanishes at K^*. Then $K^* - \overline{F} = (\overline{DF\,F} - \overline{DF}\,\overline{F})/\overline{DF}$. □

Lemma B.2 (Exact signal-to-noise in a paired design)**.** *If a scenario adds an effect η to the carry difference of a name and leaves its consensus noise unchanged (same seed), then $y^{scenario} - y^{clean} = \eta$ and the signal-to-noise $\|PL^{-1}\eta\|$ of Corollary 6.2 is known exactly, without noise. For a single template ($\eta = a\,t$) it equals $|a|\,\|\tilde{t}\|$.*

Proof. The noise cancels in the difference; for $\eta = at$, $PL^{-1}\eta = a\tilde{t}$. □

Lemma B.3 (Student calibration of the FO-mismark test)**.** *Let $\hat{\beta}_0 \sim \mathcal{N}(\beta_0, c\,\sigma^2)$ and let $\hat{\sigma}^2$ be an independent estimate with $\nu\hat{\sigma}^2/\sigma^2 \sim \chi^2_\nu$. Then $(\hat{\beta}_0 - \beta_0)/\sqrt{c\hat{\sigma}^2} \sim t_\nu$.*

Proof. Definition of the Student distribution: a standard normal divided by the square root of an independent χ^2_ν/ν. □

With dispersions estimated from n_A contributors, $\nu \approx n_A - 1$ at the short end, where the repo-level uncertainty is read; the level scale s then corrects what this approximation misses.

Remark B.4 (Multi-pillar second-order term). Proposition 4.7 extends pillar by pillar: for $V = \Phi(F_1, \ldots, F_K)$ with expiries on pillars, the dependence of the first-order total on the dividend panel is $-\sum_k \partial_k\Phi\, F_k\left[\frac{1}{2}T_k^2\Delta b_k^2 + T_k\Delta b_k\Delta D_k/(S - D_k)\right]$ plus the cross terms created by the interpolation of carry between pillars, all of second order.

C Synthetic market specification

- **Dividends.** EU stocks: annual dividend between 20 April and 30 May (35% of names also pay a November interim of 40%); US stocks: quarterly; hard-to-borrow: annual or quarterly; EU index: monthly stream with weights $(1, 2, 4, 14, 24, 18, 6, 4, 3, 4, 3, 2)$ by calendar month. Yield and growth drawn per name in the ranges of the category table.
- **Repo.** $b(T) = b_\infty + (b_0 - b_\infty)e^{-T/\tau}$ with b_0, b_∞, τ drawn per name.
- **Contributor dividend views.** $d_{ij} = d_i(1 + \sigma(t_i)[0.7\,z_j + \sqrt{1 - 0.7^2}\,e_{ij}])$ with $\sigma(t)$ affine in t (category-specific), z_j persistent, e_{ij} idiosyncratic; announced dividends (before 0.15 year) have one tenth of the dispersion.
- **Contributor repo views.** $b_j(T) = b(T) + \sigma_b(u_j + 0.6\,v_j\,\tau(T))$, with $\tau(T)$ a tilt linear in $\ln T$ from -0.5 to 0.5; σ_b is 15, 10, 60 and 8 bp for the four categories.
- **Options.** Strikes 80–120% of spot; volatility with a term structure and a skew of -0.10 per unit of log-moneyness, contributor volatility noise of one point, price noise 2×10^{-5} of spot.
- **Consensus.** Mean over the panel after removing quotes more than four scaled median absolute deviations from the median.
- **FO.** Dividends with 1% noise, repo with a 3 bp level error; spot equal to the Totem reference spot.

D Reproducibility

All numbers, tables and figures of this note are generated by the repository:

```
pip install -e .[benchmark,report,dev]
pytest
python scripts/run_benchmark.py    (about 4 minutes)
PYTHONPATH=.  python scripts/build_latex.py
```

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